Remote Work and Residential Sorting: IV Evidence From Expiring Office Leases

by

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Abstract

How has remote work reshaped residential sorting and housing demand, and what are the implications for state and local governments? To estimate causal effects, I propose a novel instrument for remote work that exploits quasi-random variation in the timing and size of office lease expirations, captured through a Bartik-style exposure measure at the residential block level. Expirations allow tenant firms to reduce office space and switch employees to remote work, generating strong first-stage effects. Remote work causes modest increases in housing and property tax expenditures in exchange for space, homeownership, and public schools, but not other neighborhood characteristics. It significantly increases migration, particularly out of cities and states that levy income taxes. At the neighborhood level, higher 2020 remote work shares cause subsequent residential turnover, demographic clustering, and property tax revenue windfalls. Taken together, the results indicate that remote work induces migration consistent with Tiebout sorting, and accounts for 10% of migration since 2020. Residential choices and tax bases now depend less on employment proximity and more on affordability and tax-benefit linkage.

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1 Introduction

This paper estimates the causal effects of remote work on residential sorting, housing demand, and local public finance, exploiting quasi-random variation in the timing and size of office-lease expirations. Starting with the Tiebout (1956) model of "voting with your feet," a classic question in labor, public, and urban economics is how households choose where to live and how this affects local governments. Traditionally, commuting costs anchored households near workplaces, but the widespread adoption of remote work has relaxed this constraint, allowing households to choose locations that better reflect their preferences for housing, amenities, and public services. In particular, remote work may shift migration away from traditional employment centers to locations with space, affordability, school quality, and tax treatment. The resulting fiscal implications have drawn substantial public attention, including high-profile disputes between states such as New York and New Jersey over how to tax remote workers, and news of population and tax base loss. Thus, remote work links residential sorting directly to the fiscal structure of local and state governments.

Despite extensive recent research and large potential effects, it is difficult to causally estimate the impact of remote work on residential choice. First, individuals self-select based on unobservable preferences, which can reflect personality characteristics, family structure, and work ethic. Second, the direction of causality is ambiguous. Namely, migration and remote work can be a joint simultaneous decision, migration can result from remote work, or remote work can result from migration. Third, there is the confounding impact of the pandemic, which coincides with the prevalence of remote work through its impacts on infection rates, social preferences, and macroeconomic factors.

I address endogeneity concerns by exploiting the inflexible timing and size of office lease expirations as a quasi-random source of variation in remote work. Lease expirations provide firms the opportunity to adjust physical office space and thus adjust remote work policies, while existing contracts prevent such changes. Office leases are long contracts lasting

3 to 15 years that lack tenant protections and carry significant financial and reputational costs for early termination. I quantify an individual's exposure to office lease expirations using a Bartik-style instrument based on their previous-year residence block. Exposure is defined as the square footage of leases expiring in the work blocks employing residents of that residence block, scaled either by employment or by time and weighted by publicly available employment flow data. Intuitively, if most residents of a block are employed in areas where many large office leases are nearing expiration, those residents are more exposed to potential remote work transitions.

The data include the restricted-access American Community Survey (ACS) from 2005 to 2023, an annual 1% sample of the US population conducted by the US Census Bureau. It is a repeated cross-section of individuals, collecting detailed demographic, economic, and residential data, along with granular location data. To construct instrument exposure, I link each 2005–2023 ACS respondent using the previous year's block-level employment flows from the publicly available LODES from 2004 to 2022. My third dataset is a database of 1.2 million corporate leases compiled by CompStak, a commercial real-estate platform, of which 673,000 are office leases. It represents roughly 20% of the US office real-estate stock, and is used to construct my measure of office lease expirations. The analysis of remote workers spans the full 2005–2023 period, although most of the impacts of remote work are concentrated in responses from 2020 onward, when remote work became widespread and the instrument has the greatest relevance.

Remote work dramatically increases migration and jurisdictional choice, raising 1-year migration by 11 percentage points (pp), and conditional on previous residence, reduces the probability of living in states and cities levying income taxes by 2.0 and 4.0 pp, respectively. At the same time, it modestly increases annual property tax payments by $431. Remote work decreases private and increases public school enrollments for children in the household, mostly through migration.

On housing demand, remote work modestly raises monthly housing expenditures by

$130 in exchange for 5% more interior space, 29% less renting, and 27% fewer non-single-family residences, all relative to their means. However, there is limited evidence of sorting based on homophily, neighborhood demographics, or income, which are proxies for amenities. Taken together, remote work induces migration to less central locations with favorable tax treatment and better public schools, but not other neighborhood characteristics.

At the neighborhood level, a higher 2020 remote-work share causes residential turnover and demographic sorting, identified using tract-level office-lease exposure in an event study. It raises cumulative outward migration by 11.5 pp and inward migration by 9.6 pp from year 2020 to 2023. This residential turnover leads to demographic sorting on proxies for public services and amenities such as age and school enrollment type, but not income. It generates a short-term property-tax windfall, likely from reassessments following home transactions. Taken together, remote work causes sorting to jurisdictions, including school districts, cities, and states, with stronger tax-benefit linkage, signaling tax base loss and pressure for tax competition.

This paper bridges four strands of research, beginning with causal evaluations of remote work. Bloom et al. (2013) and Bloom et al. (2024b) implement randomized controlled trials within firm settings, examining worker productivity, job satisfaction, retention, and promotion. Others use instrumental variables approaches for labor market outcomes, leveraging industry propensity for remote work (Bloom et al. 2024a), local internet access (Pabilonia and Vernon 2025, Boeri et al. 2022), or a firm's political donations (Kwan et al. 2022) as instruments. Relative to these, I propose office lease expirations as an instrument, featuring quasi-random variation orthogonal to industrial and local characteristics. It predicts remote work at scale and allows me to present credible partial-equilibrium estimates of remote work on migration rates, housing demand, tax bases, and public service demand. Pending administrative approval, I plan to expand these outcomes to include effects on labor supply, wages, fertility, and family formation underlying these residential moves.

The second strand of research examines migration patterns following the rise of re-

mote work. Studies such as Ramani and Bloom (2021), Althoff et al. (2022), and Gupta et al. (2022) use high-frequency migration data, including USPS address changes, cellphone mobility records, and Zillow listings, to document movement from large, high-cost urban cores toward lower-cost suburban and rural areas within the same metropolitan regions. This migration led to a narrowing of the price gap between central cities and their surrounding areas. The empirical results are largely observational, with authors arguing migration is driven by remote and hybrid workers seeking larger homes at lower costs. Relative to these, I causally identify a different mechanism, demonstrating that migration stems from preferences for higher-value neighborhood amenities and public services, not cost or space savings, with remote work instead raising housing and property tax expenditures.

The third strand of research examines which circumstances allow for migration consistent with Tiebout sorting. Rhode and Strumpf (2003) show that gradual declines in transportation costs over decades failed to produce residential sorting or fiscal differentiation predicted by Tiebout models. Later studies instead find sorting in response to sudden local or household changes: Banzhaf and Walsh (2008) show that improvements in air quality from pollution monitoring induced migration, while Farnham and Sevak (2006) find that declining schooling demand, as the youngest child turns 18, triggers moves to lower-spending jurisdictions.[1] This paper's findings are consistent with these studies, showing that the rise of remote and hybrid work represents a sufficiently large and sudden change to induce Tiebout sorting, most likely operating through commuting costs and the valuation of amenities.

The fourth and final strand examines the implications of remote work for taxation. Agrawal and Tester (2024) and Agrawal and Brueckner (2025) mention or display theoretically how remote work may alter state tax bases, with Agrawal et al. (2025) providing evidence of revenue losses in sales taxation. My contribution is to document, at a large scale, that remote work leads to migration away from states and cities that levy income taxes, which intensifies pressure for tax competition and reduces the ability of jurisdictions

[1] Saltz and Capener (2016) provide a review of additional examples.

to tax. Property taxes, however, are an exception, perhaps reflecting that they are smaller and less visible than income taxes, and have a clearer tax-benefit linkage to space and public schools.

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 presents the empirical strategy for estimating individual-level effects of remote work, including the construction of the instrumental variable and regression specification. Section 4 outlines the event-study framework used to evaluate neighborhood-level outcomes. Section 5 presents individual-level regression results and Section 6 presents neighborhood-level event study results. Section 7 demonstrates the robustness of results to alternative instrument specifications and additional controls. Finally, Section 8 concludes.

2 Data

In this section I outline the datasets used in this study, starting with the novel dataset of office lease data purchased from CompStak. I then move on to descriptions of the more familiar American Community Survey (ACS), with a discussion of outcome variable construction. Finally, I describe the public LODES data used for linkages needed for instrument construction.

2.1 CompStak

I use a proprietary database of commercial leases from CompStak to construct a measure of office lease expirations as an instrument for remote work. CompStak is a commercial real-estate platform that provides data on pricing, comparables, and property valuations to corporate clients. It collects lease information from 42,000 brokers managing 3.6 million properties nationwide.

A key advantage of the dataset is not only its broad national coverage but also the granular detail it provides for each lease transaction. The compiled and standardized

data contain nearly complete information on the size, duration, start date, end date, and coordinates of each lease, which I use to measure annual lease expirations at the block level.[2] In addition, it includes, with varying degrees of coverage, rich data on building type, building class, tenant name, tenant industry, and current operating status. However, without additional linking, it does not report detailed information on lease or tenant status at or after expiration, leaving the ultimate renewal and firm migration outcomes unknown.

The dataset I obtained contains 1.2 million leases across all 50 states and every major US city, with geographic coverage shown in Figure 1. It includes 673,000 office leases spanning 2.8 billion square feet, approximately 20% of the US office real-estate stock as measured by the Department of Energy. Coverage is strongest in large urban counties but exhibits substantial variation both across and within urban and rural areas. Office leases make up the majority of transactions (57%), with industrial (23%) and retail (17%) properties accounting for most of the remainder. Among office leases, 52% are new leases, 26% are renewals, 5% are expansions, and 17% are of unknown type. Fewer than 0.1% are restructures, which include cases in which tenants renegotiate their leases to reduce cost or duration.

The annual coverage is shown in the left panel of Figure 2, with most lease start dates between 2000 and 2025. Coverage gradually increases beginning in 1990 and peaks in 2019, immediately before the pandemic. The subsequent decline in originating leases reflects both reduced demand for office space and lags in data collection updates. Frequencies are weighted by square footage to reflect variation relevant to instrument construction and to capture the fact that larger office leases employ more workers. The right panel shows lease start dates among leases expiring in 2020–2023, the years in which firms were most likely to leverage expirations to downsize or transition to remote work. Of these, 76%, representing 90% of the square footage, are contracts originating in 2019 or earlier. Figure A1 presents an unweighted version, showing a similar trend.

[2]I define blocks specifically as Census blocks, the smallest geographic unit used by the US Census Bureau, containing on average 41 residents. They typically correspond to a city block bounded by streets or other visible features, though they can cover larger areas in suburban and rural regions.

Figure 3 displays typical lease durations. Most leases are multi-year contracts, with larger leases tending to have longer durations. The left panel, unweighted by size, shows that the most common durations are 5-, 3-, and 1-year contracts. When weighting by size—reflecting the offices that likely employ the most workers—the duration distribution shifts upward, with an average of 8 years and most leases ranging from 3 to 15 years. Finally, Figure 4 shows the distribution of lease sizes, which features a long right tail. Most leases are under 10,000 square feet, though these represent only a small share of total office space. As shown at the bottom of the right panel, the average employee works in an office covering 83,187 square feet.

2.2 American Community Survey

I use the large, comprehensive, American Community Survey (ACS) from the US Census Bureau to determine remote-work status and outcome indicators of Tiebout sorting. In particular, I utilize a restricted-access version covering 2005 through 2023, which in addition to the demographic, economic, and residential variables in the publicly available version, includes granular block-level location of place of work, residence, and previous year residence. This detailed location data allows for granular block-level instrument construction, detailed outcomes on migration and tax sorting, and measures of neighborhood characteristics.

The rate of remote work amongst employed individuals over time is shown by Figure 5. I classify individuals as working remotely if they self-report "worked at home" as their typical mode of transportation to work in their ACS responses. Since the ACS does not ask about hybrid work arrangements, this measure likely reflects workers whose jobs are predominantly or entirely remote.[3] Based on this definition, the remote-work share gradually grew from 4% in 2005 to 6% in 2019, sharply increased to 15% in 2020, peaked at 19% in 2021, and dropped to 14% in 2023. Evidence from Barrero et al. (2023) and U.S. Bureau of Labor Statistics (2025) suggests that the rate of remote work, albeit defined more broadly, has been stable

[3]This is further supported by analysis in Buckman et al. (2025).

since then.

Although most of remote work's effects are concentrated in the 2020–2023 period, having data starting in 2005 offers several advantages beyond simply increasing statistical power. The longer panel enables interpreting results as the general impact of remote work on residential sorting, rather than a narrow interpretation tied only to its post-pandemic increase. This is particularly useful because, despite lower overall prevalence before 2020, 53% of remote workers in the public ACS appear in those earlier years. Finally, an extended time series also facilitates credible pre-trend analysis in neighborhood-level event study regressions.

Migration and tax-sorting outcomes are defined using respondents' current and previous-year block-level residences.[4] With this information, I construct a migration indicator denoting whether a respondent changed residence within the past year. The detailed location information further allows identification of whether the respondent currently lives in a state or city with an income tax, or previously did so within the past year, as well as the direction of any migration between such jurisdictions. Currently, nine states do not levy a personal income tax: Alaska, Florida, Nevada, New Hampshire, South Dakota, Tennessee, Texas, Washington,[5] and Wyoming. In addition to sales and property taxes, several major cities also impose local income taxes. There are also thousands of smaller taxing jurisdictions, such as all counties in Indiana and hundreds of municipalities in Iowa, Kentucky, and Ohio. Notable examples include New York City, Newark, Philadelphia, Pittsburgh, Washington, D.C., Detroit, St. Louis, Kansas City, Birmingham, and San Francisco. The final tax outcome I consider, which overlaps with housing costs, is annual property taxes paid by the respondent's household, with renters coded as 0.

Outcomes measuring demand and utilization of public services focus on workers' school enrollment choices for their children. Using household identifiers, I identify whether

[4]Specifically, the ACS asks, "Did this person live in this house or apartment 1 year ago?" and "Where did this person live 1 year ago?" with follow-up questions on street address, city, county, state, and ZIP code.

[5]Washington State has a narrow capital gains tax but no broad-based income tax.

workers live with children aged 3–17 enrolled in public or private school. The omitted category includes respondents without school-aged children and, to a lesser extent, those with homeschooled children. Because the analysis conditions on employment, the sample is restricted to respondents aged 16 and older, so these variables primarily capture the enrollment decisions of parents, guardians, and other relatives.[6]

Since the ACS does not record prior-year school enrollment, I approximate school-switching behavior by combining indicators for migration and for whether a household has children in public or private school. If remote workers relocate to a preferred school district for the purpose of shifting their children from private to public schools, they would appear as households that both migrated and have children enrolled in public school, but not in private school. Comparing how often these patterns occur helps identify whether households are substituting through migration.

To measure housing expenditures, the ACS contains household-level statistics on monthly housing costs (includes rent and mortgages), home values (renters coded as 0), and ownership status[7]. It also reports detailed housing characteristics, which I use to examine outcomes such as the number of bedrooms, total rooms, housing type, and lot size.

Finally, to examine sorting on neighborhood characteristics, I use detailed Census block-level data on previous and current residences to construct neighborhood measures defined at the Census tract level. For individual-level analysis, this allows me to identify whether workers with children live in neighborhoods with more children, whether individuals reside in areas matching their own race or ethnicity, and how their income compares with the average income of their current and previous neighborhoods. For neighborhood-level analysis, this enables calculation of demographic heterogeneity measures within neighborhoods.

[6]The ACS records parent–child relationships only for the household head, but for most working adults in family households, the presence of children aged 3–17 can be a strong indicator of parent or guardian status. According to Chetty et al. (2020), 93% of children identified through tax returns lived in the same household as their parents.

[7]It only indicates whether the household owns or rents, but not the specific individuals owning or leasing.

2.3 LODES

The Longitudinal Employer-Household Dynamics (LEHD) datasets from the US Census Bureau use state unemployment insurance records to build various datasets regarding employment. Among these, the LEHD Origin-Destination Employment Statistics (LODES) dataset reports the number of workers living and working in every combination of Census residence block and work block.[8] This dataset thus identifies the set of potential work blocks associated with a respondent's current and previous-year residence block, and assigns corresponding probabilities to each possible workplace block location.

3 Empirical Strategy for Individual Outcomes

The goal of this study is to estimate how remote work affects individual outcomes related to residential choice. Specifically, I examine the dimensions along which individuals adjust their housing and location decisions—including choices of homes, neighborhoods, schools, and jurisdictions—once traditional commuting constraints are relaxed. I begin by describing institutional details of office leases, which provide the source of quasi-random variation in remote work. I then construct a Bartik-style exposure measure at the residential block level. Finally, I outline a model linking individual residential and jurisdictional choices to remote work.

3.1 Office Leases

I leverage the timing, size, and location of lease expirations as a quasi-random source of variation for remote work. Much attention has focused on how the increase in remote work has significantly decreased demand and increased vacancy in many central business districts.

[8] In LODES, both "residence" and "work" blocks refer to Census blocks, the smallest geographic unit used by the US Census Bureau, often corresponding to city blocks bounded by streets. A block can appear as either a residence or a workplace (or both), depending on whether it contains homes, offices, or both.

However, there is now limited but growing discussion of the opposite mechanism I utilize, that an expiration increases a company's ability to adjust office space demand downward, and thus facilitates adoption of remote policies for its employees.

A 2023 Goldman Sachs report mentions that companies locked into long-term leases can typically reconsider their office needs only at expiration. A report by Resume.org (2025), which gained some media attention (e.g., Fortune articles Morse (2024), Thier (2024)) indicates that 33% of companies report reducing remote work due to existing office lease agreements. This can be attributed to perceived benefits of in-office work, or to sunk-cost motivation to use existing office space. Conversely, 10% of firms stated that they would use expirations as a chance to downsize and transition to remote policies.

Unlike residential leases, commercial leases offer few statutory tenant protections and impose substantial penalties for early termination. Ending or downsizing a lease before expiration typically incurs large fees specified in the original contract, often exceeding 50% of the remaining rent. Smaller leases tend to face lower percentage penalties, while large leases covering a substantial share of a building can face fees as high as 90%. In addition to these financial penalties, early termination can entail significant reputational costs. Because brokers and landlords frequently share information and serve as tenant references, a firm that breaks its lease may face greater difficulty securing future office space.

Figure 6 presents a hypothetical example based on a common 5-year lease term with an early termination penalty equal to 50% of the remaining rent. It also incorporates reputational and fixed termination costs totaling 20% of the lease's total value. The figure illustrates conceptually that early termination costs are substantial and decline only gradually as the expiration date approaches, remaining significant even near the end of the lease term. In accordance with this, lease restructures are relatively uncommon, occurring in less than 0.1% of office leases in my dataset and also shown to be infrequent in prior academic research (e.g., Fisher and Ciochetti 2007). Thus, observed end dates represent binding commitments that firms cannot easily endogenously alter to downsize or shift to remote work,

making lease expirations reasonably treated as predetermined and exogenous to a firm's short-term strategic choices.

The first key source of variation, and the most plausibly quasi-random, is timing. Consider two characteristically similar Firms 1 and 2, both in the same sector with a high propensity for remote work, led by managers with potential concerns about workplace culture and productivity resulting from remote work. Both firms have 5-year office leases, Firm 1 signing in 2014, and Firm 2 signing in 2016. Firm 1 renews its lease in 2019, extending it to 2024. Starting in 2020, both firms see a sharp increase in hybrid or fully remote work. In 2021, when remote work is at its peak, and despite potential managerial concerns regarding remote work, Firm 2 can benefit from retaining or increasing its hybrid or fully remote work arrangements by using the opportunity to reduce office demand. However, Firm 1 does not receive the same cost-saving benefit of retaining or increasing hybrid or remote arrangements, since it is locked into a lease expiring in 2024, with a very costly exit fee. Since it has the locked-in cost of an office lease anyway, managers can require a return to office to avoid wasting space while regaining in-person benefits.

A second key source of variation is the scope of leases across both location and size. Since lease expirations vary by tenant, this generates substantial variation between blocks within central business districts and outlying employment centers. Using public data on employment flows, which identify blocks with the most residents employed in work areas with expiring leases, this further translates into variation in exposure between blocks within residential areas.[9] Finally, larger leases both involve more employees and impose stronger constraints or incentives on firms' remote-work decisions.

3.2 Lease Expiration Measure

I construct an annual, block-level Bartik-like measure of exposure to expiring office leases, based on each respondent's previous-year residential block. Because I cannot identify spe-

[9]In principle, variation in expirations by tenant firm creates variation at the worker-level based on employment relationships. However, such linkages are not observed.

cific respondent-employer relationships in the ACS, this exposure is meant to represent the expected size of office lease expirations affecting an individual's employer and entering into the employer firm's remote policies affecting the worker.[10] The larger the expiring lease, and the more concentrated the set of potential work locations, the more likely it is that a lease expiration influences an individual's likelihood of remote work by prompting changes in their employer's office policies.

Figure 7 provides an example, using lease data from CompStak and public employment flows from LODES. For the 6 residents identified to live in the red highlighted residence block in Manhattan, I identify 5 potential work blocks, each with office lease expirations that year ranging from 0 square feet to 283,972 square feet. The exposure measure is thus the average square footage of these expirations in these 5 work blocks, weighted by employment shares; 4 of the 5 work blocks each employ 1 resident, and are thus assigned a weight of $\frac{1}{6}$. One work block employs 2 residents, and is thus assigned a weight of $\frac{2}{6}$.

Formally, an individual's exposure to office lease expirations is defined as the following weighted average:

$$expsqft_{rt} = \sum_{w \in W_r} \frac{emp_{w,r,t-1}}{emp_{r,t-1}} expsqft_{wt} \tag{1}$$

The left-hand side term varies by current year t and previous-year residence block r, with all ACS respondents in the same block and year sharing the same block-level exposure. The right-hand side term $expsqft_{wt}$ is the square footage of office space leases that is up for renewal in a work block w during year t. I then take the weighted average across all work blocks in set W_r, the set of work blocks w which employed residents of r in year $t-1$. The weights are lagged employment shares $\frac{emp_{w,r,t-1}}{emp_{r,t-1}}$ for each block from LODES, where $emp_{w,r,t-1}$, the number of residents of block r employed in block w during year $t-1$, and $emp_{r,t-1}$ is the total number of employed residents of block r during year $t-1$.

[10] The restricted-access ACS lists block location of a respondent's employer, but only for those not working from home. Remote workers have work locations set equal to their home locations.

I use residence block r and weights $\frac{emp_{w,r,t-1}}{emp_{r,t-1}} expsqft_{wt}$ corresponding to $t-1$ primarily because remote work often induces migration, and current residence may not reflect their initial exposure to office lease expirations. In addition, an individual's remote-work status can conceptually be dependent on exposure and firm decisions in the previous year.

With the definition of expirations exposure in Equation 1, I define 3 more variations of the instruments I use in my analyses. The second variant is a 5-year cumulative measure across years $t-4$ to t. This leverages the fact that expirations in previous years, and not only in the current year, may have led employers to adopt remote policies.

$$expsqft_{rt}^{(5)} = \sum_{s=0}^{4} expsqft_{r,t-s} \tag{2}$$

The third variant normalizes by employment sizes of blocks, which changes the measure from a measure of expiring square feet to expiring square feet per worker. This instrument therefore captures the size of expiring leases relative to the local workforce rather than in absolute square feet.

$$expnorm_{rt} = \sum_{w \in W_r} \frac{emp_{w,r,t-1}}{emp_{r,t-1}} \frac{expsqft_{wt}}{emp_{w,t-1}} \tag{3}$$

The term $expsqft_{wt}$ for expiring square feet in a work block w is normalized by $emp_{w,t-1}$, the number of people working in w.

The fourth variant takes the 5-year cumulative sum of the normalized measure

$$expnorm_{rt}^{(5)} = \sum_{s=0}^{4} expnorm_{r,t-s} \tag{4}$$

For each measure listed in Equations 1, 2, 3, and 4, I also interact with indicators for calendar year. This accounts for the fact that the impact of office lease expirations on remote work can vary by year, and has the greatest positive predictive strength starting in 2020.

Due to the higher power from using the cumulative measures, the flexibility of year in-

teractions, and the more easily interpretable units of the normalized measure, I use Equation 4 interacted with calendar year as the baseline instrument functional form for my analyses, presenting others as robustness checks.

3.3 IV model for individual outcomes

I model how remote work causes individuals to sort residentially using the following equation:

$$Y_{irt} = \beta \mathbf{1}\{remote\}_{irt} + X_{irt} + O_{rt} + \alpha_R + \delta_t + \varepsilon_{irt} \tag{5}$$

The corresponding first-stage equation uses the 5-year normalized expiration exposure measure defined in Equation 4, interacted with calendar year:

$$\mathbf{1}\{remote\}_{irt} = \gamma_t expnorm_{rt}^{(5)} \times \mathbf{1}\{t\} + X_{irt} + O_{rt} + \alpha_R + \delta_t + \varepsilon_{irt} \tag{6}$$

In both equations, i indexes the individuals, r is the residence Census block during the previous year, R is the county and city of residence during the previous year,[11] and t is the year. The key explanatory variable $\mathbf{1}\{remote\}_{irt}$ is an indicator that someone self-reports working from home. Considering that remote work is a characteristic of one's employment, it is set to missing if a respondent is not employed, and the regression only includes employed individuals. Given that the analysis conditions on employment status, the term X_{irt} includes controls for detailed employment status among employed individuals (civilian, military, and employed but not currently working). Because some outcomes pertain to children in the household (school enrollment choices and neighborhood choice), X_{irt} also includes an indicator for having children aged 3–17 in the household. I include individual-level demographic and industry fixed effects as a robustness test in Section 7.2.

Many of the outcomes of interest, such as housing, residence, and migration decisions,

[11] Census designated places (CDP), which are unincorporated communities within counties, are treated as cities for fixed effects and clustering. Unincorporated non CDP areas, such as rural areas within each county, are grouped into a single category.

reflect changes relative to the previous year following a shift into remote work, making it important to condition on the characteristics of the individual's prior location. Accordingly, the term X_{irt} also includes linear controls for previous year neighborhood average income and housing characteristics such as monthly average costs, average home values, homeownership rates, the proportion of detached single family homes, the average number of bedrooms and rooms, and the proportion of homes with lots exceeding one acre. To further condition on prior year place of residence, the regression also includes fixed effects α_R for the previous year city and county of residence. Finally, I include δ_t fixed effects for calendar year.

To mitigate omitted variable bias, I also include in both equations O_{rt} containing several controls to account for office real-estate characteristics of the work blocks W_r employing residents of block r. The first control is a quadratic in the total amount of active office space in those work blocks appearing in CompStak, weighted by employment shares. This accounts for the fact that work blocks with larger office stocks are often located in central business districts and may employ more individuals capable of working remotely. The second control is a quadratic in the average duration of leases in the same work blocks, since shorter leases imply higher turnover and more expirations but may also reflect smaller firms with different remote-work propensities. The third is the total square feet of office space in the counties employing residents of r, based on Department of Energy (DOE) tallies. The fourth is the sample coverage ratio, the share of total office space in DOE data that is represented in CompStak, accounting for potential impacts of sample coverage not captured by existing fixed effects. As a robustness check and following Borusyak et al. (2025), I control for the expected value of the instrument, expected annual expirations, approximated as total leased square footage divided by average lease duration.

4 Empirical Strategy for Neighborhoods

The goal of this section is to estimate how the increase in remote work reshaped neighborhoods, linking individual migrational and jurisdictional sorting to neighborhood-level outcomes, including residential turnover, demographic sorting, changes in public service demand, and changes in government revenue. I start by extending the Bartik-style exposure measure used for individuals (Equation 1) from the residential block level to the Census tract level. I then outline an instrumented dynamic difference-in-differences regression design.

4.1 Lease Expiration Measure for Neighborhoods

I define the unit of analysis as the Census tract, the Census Bureau's standard neighborhood unit, with an average population of roughly 3,900 residents and containing approximately 100 Census blocks (U.S. Census Bureau 2020). Tracts are subunits of counties, with boundaries often following roads, geographic features, and municipal boundaries. Boundaries always follow county lines but do not always coincide with city limits. Using information on block and tract of residence for ACS respondents, I construct a panel of Census tracts and their demographic characteristics. This allows for neighborhood-level fixed effects to account for characteristics of neighborhoods which are relatively unchanging over time.

However, it does not account for potential confounding from factors varying both across time and location, including from the pandemic and other macroeconomic shocks. I thus aggregate the block-level office lease expiration instrument in Equation 2 to the tract-level. To start, I first aggregate $expsqft_{rt}$ from Equation 1 from block r up to its tract k as the following weighted average

$$expsqft_{kt} = \sum_{r \in R_k} \frac{emp_{r,t-1}}{emp_{k,t-1}} expsqft_{rt} \tag{7}$$

where R_k is the set of residence blocks r in tract k, and the fraction $\frac{emp_{r,t-1}}{emp_{k,t-1}}$ is the

employment share of block r within tract k. I then take the 5-year sum and interact with calendar year

$$expsqft_{kt}^{(5)} \times \mathbf{1}\{t\} = \sum_{s=0}^{4} expsqft_{k,t-s}\mathbf{1}\{t\} \tag{8}$$

With this, some neighborhoods can be instrumented to have higher remote work shares at the onset of the remote work trend. Similar to the more granular block-level instrument in Equation 2, the more residents of tract k who are employed in blocks $W_{k,t-s}$ with large expiring leases, the more the tract k is exposed to office lease expirations.

4.2 Dynamic DiD for Neighborhoods

The mechanism of interest is whether greater remote work leads to increased mobility and, in turn, selective exit and replacement of residents. I estimate the following instrumented dynamic difference-in-differences (DiD) model, conceptually similar to an instrumented event study. Neighborhoods with above-average remote-work shares in 2020 form the treatment group, and those with below-average shares serve as controls. The model compares their subsequent demographic and fiscal trajectories to estimate how remote work reshaped neighborhood composition and local public finance. The estimating equation is

$$Y_{kt} = \sum_{s \neq 2019} \left[\beta_s \cdot \mathbb{1}\{remote_k > \overline{remote_k}\} \times \mathbf{1}\{s\}\right] + \delta_t + \alpha_k + O_{kt} + \epsilon_{kt} \tag{9}$$

where k indexes Census tracts, t indexes years from 2005 to 2023, α_k are tract fixed effects, and δ_t are year fixed effects. The key explanatory variables are the event dummies $\mathbb{1}\{remote_k > \overline{remote_k}\} \cdot \mathbf{1}\{s\}$. Within this, the term $\mathbb{1}\{remote_k > \overline{remote_k}\}$ indicates that a tract had above-average remote work among its residents in 2020, denoting the treatment group, and is fixed across all years t for the same tract k. The term $\mathbf{1}\{s\}$ is an indicator for calendar year within the summation. The omitted baseline year is 2019, immediately

preceding the rise in remote work in 2020. Finally, the term O_{kt} is a set of real-estate controls defined at the tract level, analogous to the term O_{rt} included in Equation 5.

Since there are 18 event dummies for years 2005 through 2023, excluding 2019, there are 18 of the following first stages:

$$\mathbb{1}\{remote_k > \overline{remote_k}\} \times \mathbf{1}\{t\} = \sum_{s=2005}^{2023} \gamma_s \left[expsqft_{ks}^{(5)} \times \mathbf{1}\{s\} \right] + \delta_t + \alpha_k + O_{kt} + e_{kt} \quad (10)$$

for each $t \in \{2005, ..., 2023\} \setminus \{2019\}$, using the instrument defined in Equation 8.

5 Results for Individuals

I begin by presenting first-stage results that demonstrate the strength of the instrument. I then discuss second-stage results for individual-level outcomes, starting with public finance and tax migration, and proceeding to changes in education demand, housing demand, and neighborhood amenities. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county.

5.1 First Stage

To estimate the effect of the lease-expiration instrument on remote work, I report first-stage regressions based on Equation 6 using alternative formulations of the Bartik-style exposure measure defined in Equations 1, 2, 3, and 4, along with versions interacted with calendar year. Results are shown in Tables 1, 2, 3, and 4.

Following the order of instrument construction, I first present in Table 1 Column (1) the current-year measure of expirations $expsqft_{rt}$ (Equation 1), and in Column (2) the same measure interacted with calendar year. Exposure to current-year expiring leases, constructed as a Bartik-like measure at the Census-block level, has strong predictive power for individual remote work. For the average worker in an 83,187-square-foot[12] office, an expiration leads

[12]This is the size-weighted average, which better weights by employment and represents the size of the

to a small but statistically significant 0.66 percentage point (pp) increase in the probability of remote work, with an F-statistic of 38.69.

Interacting with calendar year reveals that the coefficient in Column (1) combines opposing effects before and after 2020. Beginning in 2020, when remote work rose sharply, I observe evidence that employers started using lease expirations as opportunities to downsize and transition to remote arrangements, while this is absent prior to 2019. One plausible explanation is that during the earlier period of economic expansion, lease expirations more often led firms to renew or relocate to slightly larger offices, maintaining office employment and marginally reducing remote work. The smaller and less precisely estimated coefficients in these years reflect both the weaker mechanism and limited data coverage. At the peak of remote work in 2021, the estimated effect implies that an 83,187-square-foot lease, the typical lease for the average worker, corresponds to a 7.1 pp increase in remote work. For large leases at the 90th and 95th percentiles of the size-weighted distribution (214,000 and 350,000 square feet), expirations increase remote work by 18.3 pp and 30.0 pp, respectively. The joint F-statistic is 29.72.

Moving to Table 2, Column (1) presents the five-year cumulative measure of expirations, $expsqft_{rt}^{(5)}$ (Equation 2). Following the same structure as the previous table, Column (2) reports the same measure interacted with calendar year. Comparing Column (1) in Table 2 to Table 1, the coefficient magnitude increases by 41%, accompanied by a substantial rise in statistical significance ($F = 93.67$). Including calendar-year interactions produces the same pattern, albeit with smaller magnitudes and slightly higher precision ($F = 38.49$). For the same 83,187 square-foot expiration, the pooled coefficient in Column (1) implies a 0.9 pp in remote work, while at the 2021 peak this effect rises to 3.7 pp. For large leases at the 90th and 95th percentiles of the size-weighted distribution (214,000 and 350,000 square feet), expirations increase remote work by 9.6 and 15.8 pp, respectively.

Tables 3 and 4 present analogous results for the measures normalized by employment

typical lease employing the typical worker.

size of the work block from Equations 3 and 4. The normalized versions, especially when not interacted with calendar year, have lower statistical significance than the preceding versions, with first-stage F-statistics ranging from 8.382 for the annual version without calendar year interactions, to 57.65 for the 5-year version without calendar year interactions. From Table 3, we see that 100 square feet of expiring office space per worker in 2020 increases the probability of remote work by 3.5 pp.

The 5-year normalized specification interacted with calendar year, shown in Table 4, is used as the baseline instrument for the second stage for several reasons. First, its first-stage F-statistic of 14.55, while not the highest among the alternatives, is sufficient for reliable inference. Second, normalizing by employment size converts the instrument into expiring square feet per worker, yielding coefficients with a clear and interpretable scale. Third, this normalization better reflects the economic scale at which office lease expirations plausibly affect remote work propensities. Fourth, interacting the instrument with calendar year allows the magnitude and direction of effects to vary over time, particularly before versus after 2020. Finally, using a five-year cumulative measure increases statistical power. Section 7.2 demonstrates robustness to alternative instrument specifications. Qualitative results are unchanged across instrument functional forms, particularly among specifications that include required year interactions.

5.2 Tax Sorting

I begin by examining how remote work affects where individuals choose to live, focusing on its role in driving migration across fiscal jurisdictions. Table 5 presents the first results, showing that remote work significantly increases migration, raising 1-year migration rates by 11.3 pp relative to an average rate of 9.6% within the sample and 13.1% in the entire public ACS.

Given the large increase, I first assess the fiscal consequences of remote work, testing whether it drives tax-motivated migration and limits the capacity of state and local govern-

ments to levy income taxes.[13] I find that, conditional on previous-year residence, remote work decreases the probability of living in a state with an income tax by 2.0 pp and in a city with an income tax by 4.0 pp. Both effects are large relative to the baseline shares of individuals living in such jurisdictions and indicate reduced capacity for state and local governments to tax income of their residents. As two important caveats, state income tax migration patterns may also reflect political preferences, as most states without income taxes, especially in recent years, lean Republican or are politically mixed. City income tax migration can also reflect the trend of remote workers exiting major cities. Appendix Table A1 presents alternative measures of this outcome, using indicators for moves into and out of taxing jurisdictions, documenting that remote work substantially increases migration out of income-taxing cities and states, inducing little to no offsetting inward migration.[14]

These patterns of tax-motivated migration motivate examining how remote work affects fiscal outcomes at more granular local levels, including city, school district, and neighborhood revenues, through property tax payments. Remote work modestly increases household property tax expenditure by $431.10 per year. Although property taxes are often embedded in overall housing expenses and often paid indirectly through mortgages, this finding suggests that remote work increases expenditures on housing-related and neighborhood-level taxes tied to school districts and local services. Taken together with the results on income-tax migration, these patterns imply that remote work shifts tax capacity from broad-based income taxation toward property taxation, reflecting a lower willingness to pay for taxes with weak perceived benefit linkage and a higher willingness to pay for local taxes tied more directly to visible neighborhood services and amenities.

[13]These results are preliminary and do not yet incorporate the full list of smaller taxing jurisdictions.

[14]To maintain consistent observation counts across regressions to simplify Census review, I include all individuals regardless of state or city of residence. I condition tax variables on fixed effects for previous-year city and county of residence.

5.3 Education Demand

To assess whether this increase in property taxes paid is related to demand for public services, Table 6 examines indicators for whether the respondent sends their children to private and public schools.[15] Local schools are a key public service, with school quality being a major determinant of neighborhood choice for families and a large component of local public spending. Controlling for whether an ACS respondent has school-aged children, remote work reduces private school enrollment by 11.4 pp and increases public school enrollment by 6.5 pp, both very large increases relative to the mean.[16] [17]

Although I cannot directly observe prior school enrollment or precisely identify substitutions, I explore potential mechanisms by constructing joint migration outcomes. Column (3) reports the joint outcome of having migrated in the past year and enrolling children in private school, while Column (4) reports the analogous outcome for public school enrollment. Because remote work substantially increases migration overall, it raises both joint outcomes. However, the effect is far stronger for public-school households, with remote work more than eight times as likely to cause households to move and have children enrolled in public school than in private school.

This pattern suggests that migration is the primary channel through which remote work alters school enrollment choices. One plausible underlying mechanism is that remote workers may have previously lived in areas closer to central business districts, where public schools were perceived as lower quality, leading them to choose private options. After transitioning to remote work, these families can relocate to neighborhoods with higher-quality public schools and reduce private tuition expenses.

[15] The variable is an indicator that a child in the household attends private or public school, respectively. Since most employed individuals are adults, this largely captures enrollment decisions of parents for their children.

[16] To maintain consistent observation counts, as with the tax variables, households without school-age children are also included for the school-enrollment outcomes, with an indicator added for whether a respondent has children aged 3 to 17.

[17] The fact that the increase in public school enrollment does not fully offset the decline in private school enrollment may reflect households with children previously split between both sectors moving entirely to public schools, or a modest increase in homeschooling.

Because local school spending is closely tied to property values and local revenue bases, this reallocation toward public schooling may widen fiscal disparities across jurisdictions. Areas with higher-quality schools are likely to attract additional in-migration and resources, while those with weaker schools may experience outflows of higher-income households. These dynamics coincide with the broader pattern of migration away from income-taxing jurisdictions, which may contribute to fiscal pressures in large cities that rely more heavily on income tax revenues to fund local services and schools.

5.4 Housing Demand

Given that remote work causes workers to sort based on preferred taxes and public services, I next examine how housing choices underlie these jurisdictional moves, as location flexibility influences both where households live and the types of homes they select, as shown in Table 7.

The results show that, contrary to prior research and popular belief, remote work slightly increases housing expenditures and purchased home values. At the same time, consistent with earlier findings, workers relocate to lower cost of living areas where those expenditures translate into higher homeownership and substantially more interior and exterior living space.

In response to remote work, households increase monthly housing expenditures, including the sum of rent, mortgage, insurance, property taxes, and other expenses, by just over $130, a 7.9% increase relative to the mean. Remote work also causes workers to choose more valuable homes, with affected households reporting $43,270 higher home values, though this is only significant at the 10% level. This may partly reflect that remote work induces some high-earning renters to purchase homes, as renters are coded as zero. In line with this, remote work raises homeownership by 7.7 pp.[18]

[18]This could also capture that remote workers are moving out of homes owned by other family members and now renting on their own. The ACS only indicates whether the household owns the home it occupies, and does not necessarily identify the owner.

Given the documented increase in housing expenditures, Table 7 next examines whether this reflects a preference among remote workers for larger indoor or outdoor spaces, given the greater time spent at home. It significantly raises the probability of living in a detached single-family home as opposed to town homes, condominiums, mobile homes, and apartments. Interior living space, measured in terms of bedrooms and rooms, increases approximately 5% relative to baseline. Finally, it increases exterior space demand; remote work increases the likelihood of living on a large lot by 4.4 pp, which is substantial compared to the mean of 14.1 pp.

The results suggest that remote work leads to substitution from renting close to work towards purchasing in lower cost suburban and rural areas. They are switching from smaller spaces in multi-family units to detached single-family homes on larger plots of land featuring larger interior space.

5.5 Neighborhood Amenities

The observed changes in housing demand and location choices suggest that remote work may also influence the kinds of neighborhoods households choose. To test this, I examine whether remote work changes residential sorting by family composition and racial or ethnic characteristics. The first outcome in Table 8 is an indicator that an individual with children in their household lives in a neighborhood with an above-average number of children. This can proxy for families moving to neighborhoods with more families, given their preferences for parks, safety, and schools. Related to this, the second variable is an indicator that a respondent lives in a neighborhood with a racial or ethnic plurality matching their own. This can proxy for homophily or preferences for amenities such as cultural institutions, stores, or restaurants which are closer to the respondent's preference. For both variables, I document modest negative sorting of around 2.4 pp, suggesting that workers do not systematically consider a neighborhood's family or ethnic characteristics.

Next, I examine indicators of sorting based on neighborhood income. The outcome in

Column (3) is $(income - \overline{income_{kt}})$, the difference between an individual's own income and the average income of their neighborhood, defined as the Census tract of residence. Results show suggestive evidence that remote work leads individuals to live in neighborhoods with lower average incomes than their own, indicating that it does not promote sorting into higher-income areas. Column (4) reports results using the squared difference, $(income - \overline{income_{kt}})^2$, which captures how closely an individual's income aligns with neighborhood income. Remote work does not appear to increase such alignment, suggesting instead that remote workers are selecting into lower-income neighborhoods that may offer more desirable amenities and public services, or are sorting to areas with lower incomes reflecting lower general costs of living.

To more concretely assess whether remote workers are increasing the income gap relative to their neighbors, Columns (5) and (6) use information on previous-year residence to calculate both the current neighborhood's average income, $\overline{income_{k,t}}$, and the prior neighborhood's average income, $\overline{income_{k,t-1}}$. Column (5) reports results for the outcome $(income_{irt} - \overline{income_{k,t}}) - (income_{irt} - \overline{income_{k,t-1}})$, a difference-in-differences measure of relative income that captures changes in an individual's income gap relative to neighborhood averages between years t and $t-1$. Consistent with the results in Column (3), remote work leads individuals to select into neighborhoods with lower relative income, though both results are not statistically significant.

Column (6) extends this by examining the difference in the squared difference $(income_{irt} - \overline{income_{k,t}})^2 - (income_{irt} - \overline{income_{k,t-1}})^2$, again confirming the pattern observed in Column (4). Finally, Column (7) uses the absolute value $|income_{irt} - \overline{income_{k,t}}| - |income_{irt} - \overline{income_{k,t-1}}|$, which along with Column (4) detects some migration away from neighborhoods with comparable incomes. Taken together, the results indicate that remote workers are not sorting positively on neighborhood income. Since neighborhood income tends to correlate with many desirable amenities, these findings suggest that remote work does not increase demand for such amenities, or that income is an imperfect proxy for these pref-

erences. An alternative interpretation is that the preference for neighborhoods with lower housing costs dominates, as remote workers are moving to higher-quality homes in slightly lower-income neighborhoods, consistent with increased housing expenditures and significant increases in home size.

Taken together, the individual-level results show that remote work induces migration consistent with Tiebout sorting. It drives relocation away from jurisdictions with income taxes that have weaker perceived benefit linkage and increases utilization and willingness to pay for local public education through slightly higher housing and property tax expenditures. Workers are also seeking larger homes in lower cost areas within their existing budgets, with less regard for other neighborhood characteristics and amenities.

6 Results for Neighborhoods

To move beyond individual-level outcomes and examine how remote work reshapes communities, I next estimate neighborhood-level event-study regressions that trace changes in migration, housing, and demographic sorting around the 2020 shift toward remote work. Second-stage results are presented in Figures 8 through 14, with first-stage estimates from Equation 10 confirming that the instrument is strong, with F-statistics ranging from 23 to 42 (see Appendix Tables A7 and A8). First examining outward migration in Figure 8, neighborhoods with higher remote work in 2020 had similar outward migration trends relative to neighborhoods with lower remote work, but see a dramatic increase in outward migration starting in 2020, peaking at 6 pp in 2021. There is a corresponding increase in inward migration shown in Figure 9, though its magnitude is generally lower than outward migration, and features a less precisely measured pre-trend. These results suggest that remote work does in fact increase outward migration, triggering residential turnover as vacancies are filled by newcomers. Though annual tract-level population estimates are not available since 2020, the rates of outward and inward migration suggest a net outward movement of residents.

I next test whether residents whose demographic profiles and public service demand differ from the neighborhood majority are more likely to move out, creating vacancies filled by newcomers more similar to those who remain. In Figure 10 there is some evidence that remote work causes clustering by age in neighborhoods, but only with a delay. This somewhat mirrors results for individuals, which demonstrated weak evidence that remote work causes those with children to relocate to neighborhoods with more families with children, which would imply clustering of individuals who are middle-age or school-age. Moving to clustering on income shown in Figure 11, the results show that the measure of income inequality within neighborhoods increases, suggesting that the migration induced by remote work triggers reverse sorting by income, which is in line with individual-level results in Table 8.

Turning next to public finance, I examine whether higher remote work increases clustering by public service demand, particularly demand for public schools. I categorize households into three groups: those with children enrolled in public school, those with children in private school, and those without school-aged children. These groups are expected to have, respectively, high, moderate, and low demand for public school quality. Concentration across the three categories is measured using the Herfindahl–Hirschman Index (HHI), which ranges from 0 to 1, with higher values indicating greater concentration.

As shown in Figure 12, a high remote work share raises the HHI by about 0.02, indicating a modest but meaningful rise corresponding to up to a 10% reallocation across categories.[19] Combined with individual-level evidence of substitution from private to public schools through migration, these findings suggest that remote work contributes to households sorting into neighborhoods that align more closely with their schooling choices. The neighborhood-level results are consistent with families with children relocating to higher-quality school districts where other public-school families tend to cluster, while childless

[19]As an illustration, consider a family-oriented neighborhood with strong public schools, where 50% of workers have children in public school, 0% have children in private school, and the remaining 50% have no children in school. If remote work leads 10% of residents whose children have finished school to relocate to retirement communities, and they are replaced by households with school-age children seeking to enroll in public schools, the shares would shift to 60%, 0%, and 40%, respectively, producing a 0.02 increase in the Herfindahl–Hirschman Index.

households are moving to retirement communities or urban areas where school amenities are less relevant.

Turning to potential fiscal effects, Figure 13 shows a sharp increase in the variation of property taxes within neighborhoods. Figure 14 reveals a concurrent rise in average property taxes, though with a less credible pre-trend. Together, these patterns suggest that residential turnover led to a wave of property transactions and reassessments, producing a short-term rise in both the dispersion and level of property taxes. On this margin, remote work appears to have generated a temporary revenue windfall for local governments.

Taken together, the effects of remote work on neighborhoods mirror those observed at the individual level and highlight broader implications for local government finance. At both levels, remote work increases migration and residential turnover but has only modest effects on demographic sorting. The most consistent evidence of sorting arises from differences in public school demand. In fiscal terms, neighborhood-level results point to at least a short-run rise in property tax revenues, mirroring the result that remote work increases property taxes paid. Yet, both sets of findings suggest potential longer-run fiscal shortfalls. Although remote work spurs both outward and inward migration, the former appears to dominate.

7 Identification and Robustness

In this section I now explore identification and robustness. For these analyses, I focus on a subset of key outcomes most relevant to the results. For placebo tests, I focus on 10 key outcomes, including 5 public finance and 5 residential choice each. For robustness, given the large variety of different specifications and control sets, I focus on 4 total outcomes: migration, public school utilization, state income tax, and city income tax.

7.1 Identification

As a test of identification, I present the results of placebo tests in Tables 9 and 10, meant to test whether the instrument is systematically correlated with demographic observables. Each placebo outcome is constructed by regressing selected outcomes on observable demographic variables which are excluded from the regression, including fixed effects for age, detailed Census industry category of employment, broad race category, broad ethnicity category, education level, marital status, sex, and the presence of children in the household.

$$Y_{irt} = \theta_{\text{age}(i)} + \theta_{\text{industry}(i)} + \theta_{\text{race}(i)} + \theta_{\text{ethnicity}(i)} + \theta_{\text{education}(i)} + \theta_{\text{marital}(i)} + \theta_{\text{sex}(i)} + \theta_{\text{children}(i)} + \varepsilon_{irt}. \tag{11}$$

The placebo outcome is subsequently defined as the fitted values

$$\begin{aligned} Y_{irt}^{\text{placebo}} &\equiv \widehat{Y}_{irt} \\ &= \hat{\theta}_{\text{age}(i)} + \hat{\theta}_{\text{industry}(i)} + \hat{\theta}_{\text{race}(i)} + \hat{\theta}_{\text{ethnicity}(i)} + \hat{\theta}_{\text{education}(i)} + \hat{\theta}_{\text{marital}(i)} + \hat{\theta}_{\text{sex}(i)} + \hat{\theta}_{\text{children}(i)} \end{aligned} \tag{12}$$

Each placebo outcome is then substituted into the second-stage Equation 5 for an IV regression in place of the original outcome.

The results of the placebo tests for the 5 key public finance outcomes and 5 key residential choice outcomes are reported in Tables 9 and 10. For each placebo outcome, I compute the placebo ratio $\frac{\beta_{main}}{\beta_{placebo}}$, which compares the magnitude of the estimated effect of remote work on the true outcome to that on its placebo counterpart. The mean ratio is 40.254 and the median is 9.349, indicating that the true estimates are typically an order of magnitude larger than the placebo estimates. Half of the placebo outcomes are statistically indistinguishable from zero, and only 3 are significant at the 5% level or better despite the large sample size, with the remaining estimates small and economically negligible. Overall, all but one outcome are either statistically insignificant, economically unimportant, or associated with large placebo ratios, leaving only public school enrollment as a potential

exception.

7.2 Robustness

I first assess robustness to alternative constructions of the instrument, considering a total of 8 variants: the 4 baseline definitions in Equations 1, 2, 3, and 4, each interacted with calendar year. Table 11 reports IV estimates for migration in the past year (from Table 5), while Table 12 reports corresponding estimates for public school enrollment (from Table 6). The state and city income tax sorting outcomes from Table 5 have robustness results in Appendix Tables A2 and A3. For all outcomes, the qualitative patterns remain broadly stable across alternative instrument constructions.

Tables 11, 12, A2, and A3 have instruments following the order of instrument construction: Columns (1) and (2) have first stages matching Table 1, which uses the annual square footage instrument from Equation 1. Columns (3) and (4) have first stages matching Table 2, which uses the 5-year square footage instrument from Equation 2. Columns (5) and (6) have first stages matching Table 3, which uses the annual square footage per worker instrument from Equation 3. Columns (7) and (8) have first stages matching Table 4, which uses the 5-year square footage per worker instrument from Equation 4. For each pair, the even columns (2), (4), (6), and (8) interact the instruments with calendar year, with (8) being the baseline specification used for the results in Tables 5, 6, 7, and 8.

Qualitative results are broadly stable across the four outcomes and eight instrument variants. For both outcomes, the even-numbered columns, which interact the instrument with calendar year, deliver the most consistent and precise estimates. This reflects the fact that this functional form better captures year-specific differences in how lease expirations affect remote work in the first stage. In Table 11, remote work increases migration by roughly 11 to 16 pp relative to a mean of 10 percent, while in Table 12 it consistently increases public school enrollment by about 7 pp. By contrast, the odd-numbered columns (1, 3, 5, and 7), which omit year interactions and are therefore more likely to be misspecified, exhibit greater

variation in point estimates and substantially larger standard errors, but maintain correct direction.

The next set of robustness checks in Tables 13, 14, A4, and A5 vary the control sets and the clustering of standard errors. Following Borusyak et al. (2025), Column (1) adds a control for the expected value of the instrument, defined as expected annual expirations and approximated by $\frac{TotalSquareFeet_{r,t}}{AverageDuration_r}$ the total square footage of active leases divided by the average lease duration. Including this control has little effect on the baseline point estimates. Column (2) adds fixed effects for detailed Census industry, changing magnitudes but maintaining qualitatively similar estimates for both migration and public school enrollment. Column (3) further adds the full set of demographic controls listed in Equation 11, including age, race, ethnicity, education, marital status, sex, and the presence of children, with results relatively unchanged relative to baseline. Finally, Column (4) returns the control set to baseline as shown in Equation 5 and replaces interactive clustering with multiway clustering; although this increases standard errors, the main results remain statistically significant.

8 Conclusion

By leveraging plausibly exogenous variation in remote work induced by office lease expirations, I provide the first large-scale quasi-experimental estimates of the impacts of remote work. The findings show that remote work leads individuals to substantially adjust their residential choices, gravitating toward neighborhoods and jurisdictions offering greater perceived value in housing and public services. Remote workers substitute their children from private to public schools and are willing to bear slightly higher housing and property tax costs to do so. At the same time, they migrate away from jurisdictions with income taxes, which may be perceived as offering weaker links between taxes and benefits. The slight increase in housing expenditures in exchange for far larger living spaces and higher homeownership suggests that while workers are cost conscious, cost savings are not the primary

driver of relocation.

These results are consistent with classic models of Tiebout sorting, where greater mobility makes residents more responsive to differences in local taxes and public services. By loosening the link between where people live and work, remote work makes residential choices more sensitive to local fiscal and amenity conditions. These findings help explain post-2020 migration patterns, with the effects of remote work accounting for roughly 10% of total migration during that period. Remote work increases demand for living farther from neighbors and urban centers, with workers choosing detached homes on larger lots, even when these homes are slightly costlier. This suggests that preferential income tax treatment, quality local public services, and greater space, rather than affordability, are the dominant motivators of relocation.

For state and local governments, the results indicate that income-taxing jurisdictions can no longer rely on employment concentration or lower housing costs to retain residents. Remote work drives outward migration from these areas, reducing their ability to raise revenue. Jurisdictions with weak public schools may face longer-term tax-base erosion unless they continue to improve service quality. Together, these results point to mounting pressure for interjurisdictional tax competition.

This paper provides the first large-scale quasi-experimental evidence on how remote work reshapes residential choice and its impacts on local fiscal capacity. Building on the instrument strategy, future research could more closely examine the determinants of firms' remote work policies, including how office size, location, and amenities evolve around lease expirations before and after 2020, and how firms design incentives for employees to return to on-site work. Further work could structurally quantify changes in tax elasticities induced by remote work and evaluate the resulting general-equilibrium effects on neighborhoods, cities, and states. Finally, restricted-access LEHD linkages of individuals to employers with expiring leases would allow for a more precise identification of the causal effect of lease expirations on remote work adoption.

As extensions of this work, I conduct preliminary analyses using public ACS data with county-level location and lease expiration information. Applying the same empirical strategy, the results suggest that remote work reduces weekly hours worked at the intensive margin by about 8 hours, indicating that residential choices may partly reflect an increased valuation of leisure time. However, preliminary findings remain inconclusive regarding the effects of remote work on income and fertility and cannot yet determine how these outcomes influence residential decisions. Future extensions of this project, pending administrative approval, will investigate the impacts of remote work on labor supply, fertility, and household formation, and how these shape observed residential patterns.

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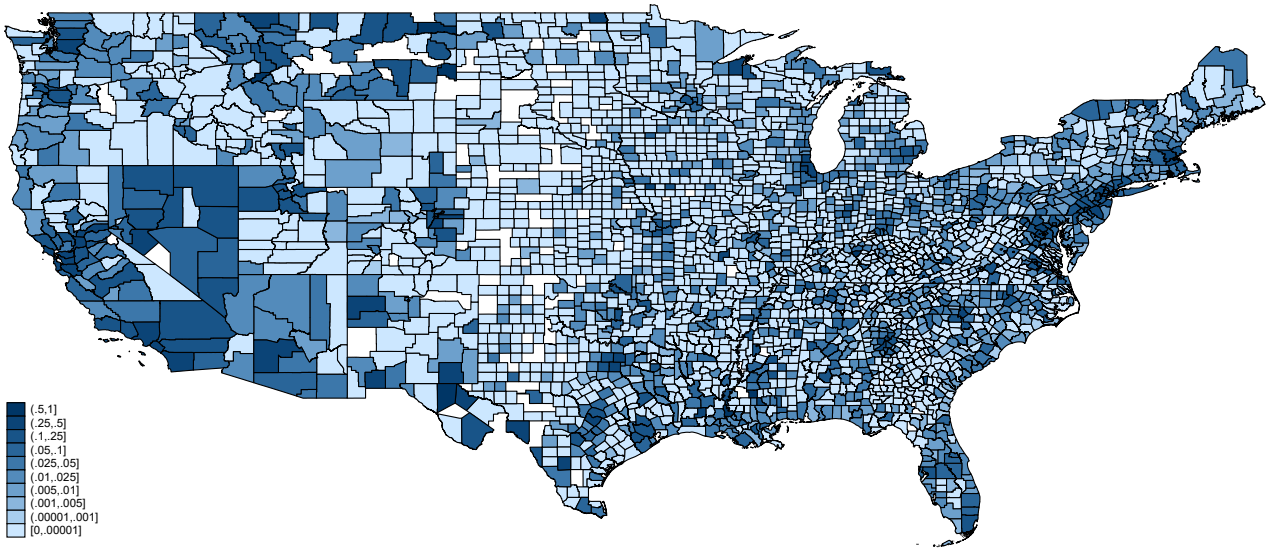
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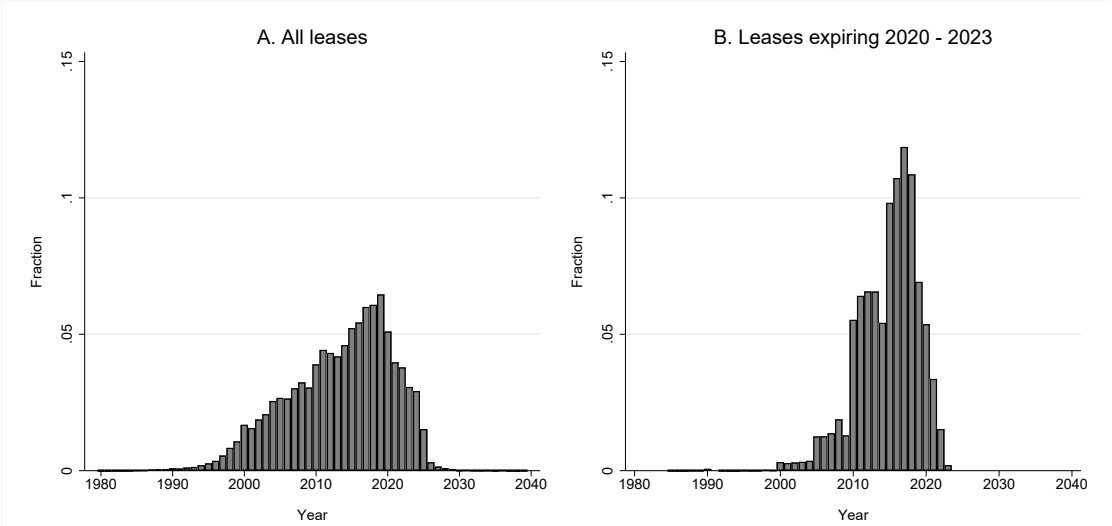
9 Figures

Figure 1: CompStak data coverage relative to total real estate stock



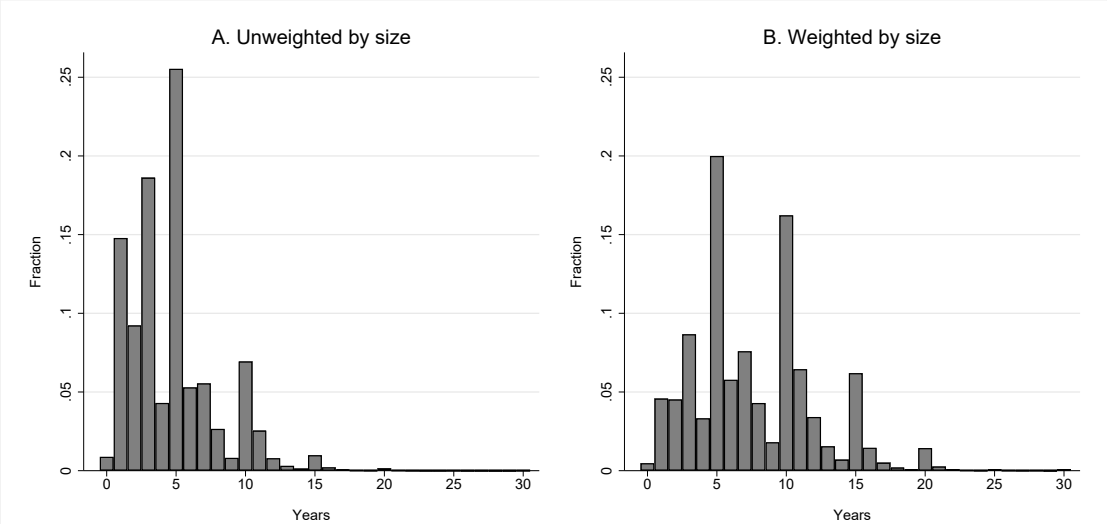
Notes: This figure shows county-level coverage of CompStak's office lease data, defined as the ratio of active office lease square footage in CompStak to benchmark tallies of office real estate stock from the Department of Energy (DOE). Coverage is calculated using each county's total square footage of office leases active in 2019, the year before the pandemic and the ensuing changes to remote work and office utilization. A lease is considered active if its term overlaps any part of 2019. Darker colors indicate higher coverage ratios. Most large urban counties, which contain most of the nation's office real estate stock, exhibit higher coverage, though many suburban and rural counties also have substantial coverage. CompStak captures 20% of total office stock as measured by the DOE.

Figure 2: Distribution of office lease square footage by start year



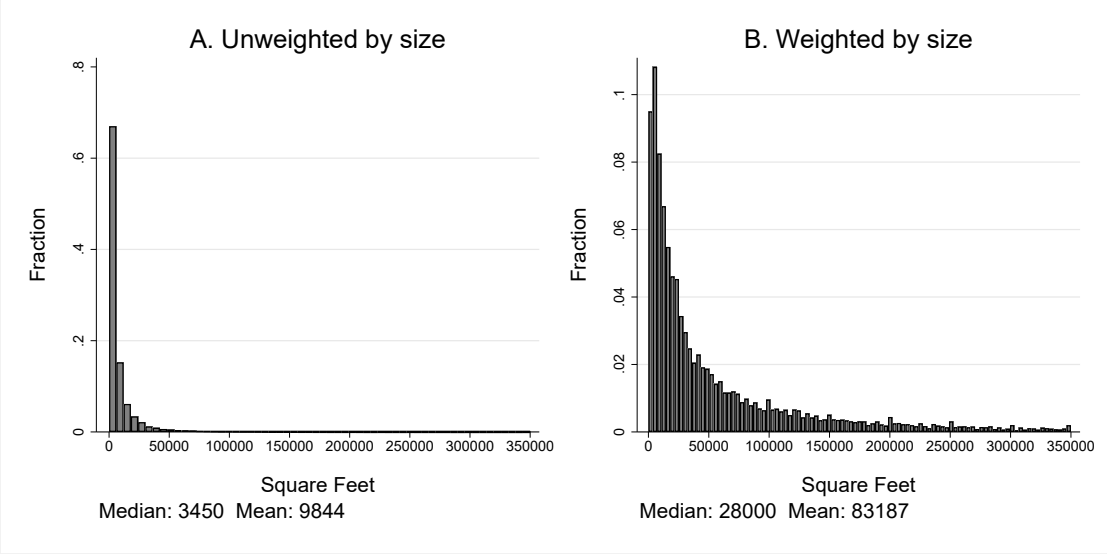
Notes: This figure shows the time coverage of the CompStak office-lease dataset. Panel A shows the amount of leased office square footage by lease start year, with both coverage and newly contracted office space peaking in 2019. A small forward-dated tail reflects leases scheduled to begin after 2024. Panel B restricts to leases expiring in 2020–2023, the contracts that provide the strongest quasi-random variation in remote work. Because large office leases have longer terms, 90% of this square footage comes from contracts signed in 2019 or earlier. Appendix Figure A1 shows the distribution of lease contracts instead of square footage.

Figure 3: Lease duration



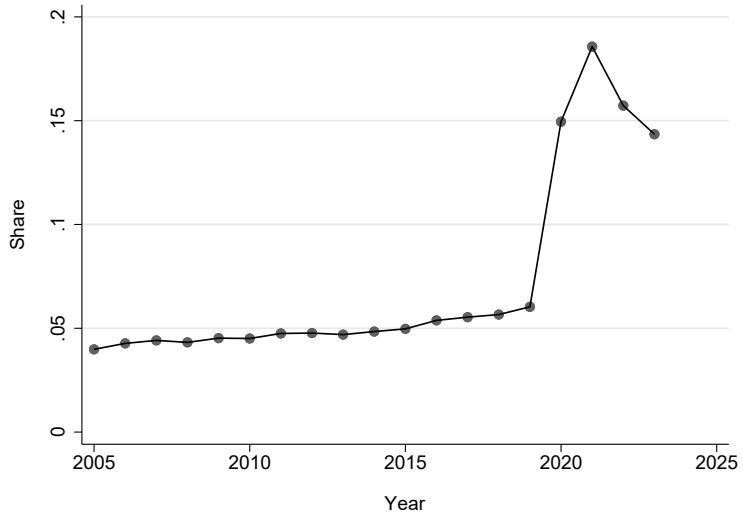
Notes: This figure shows the distribution of office lease durations in the CompStak dataset. Panel A plots lease durations in years, treating each contract as an observation. Panel B plots the same distribution weighting each lease by square footage, effectively treating each square foot of office space as an observation. Under equal weighting, the median and mean lease duration are 5 years, while under square-footage weighting they are 7 and 8 years, respectively.

Figure 4: Lease size



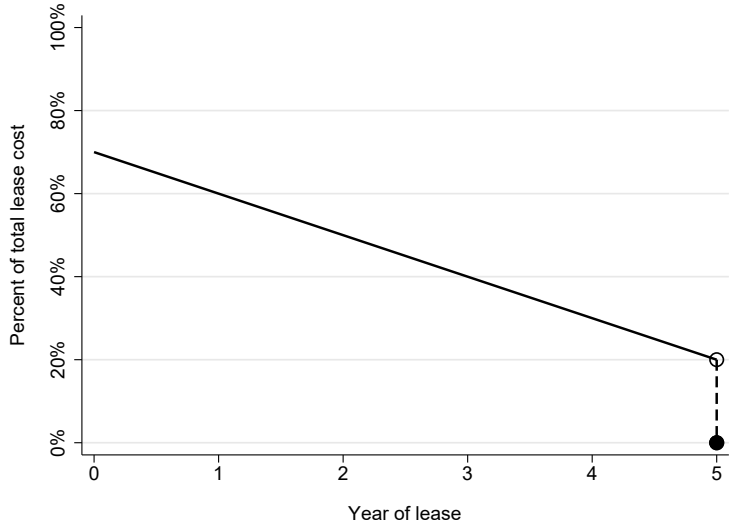
Notes: This figure shows the distribution of office sizes in the CompStak dataset. Panel A plots square footage, treating each lease contract as an observation. To reflect employment exposure, Panel B plots the same distribution weighting each lease by square footage, effectively treating each square foot of office space as an observation. Both panels are truncated at 350,000 square feet, which corresponds to the 99.9^{th} percentile of unweighted leases and the 95th percentile of the size-weighted distribution.

Figure 5: Work-from-home rate among employed ACS respondents



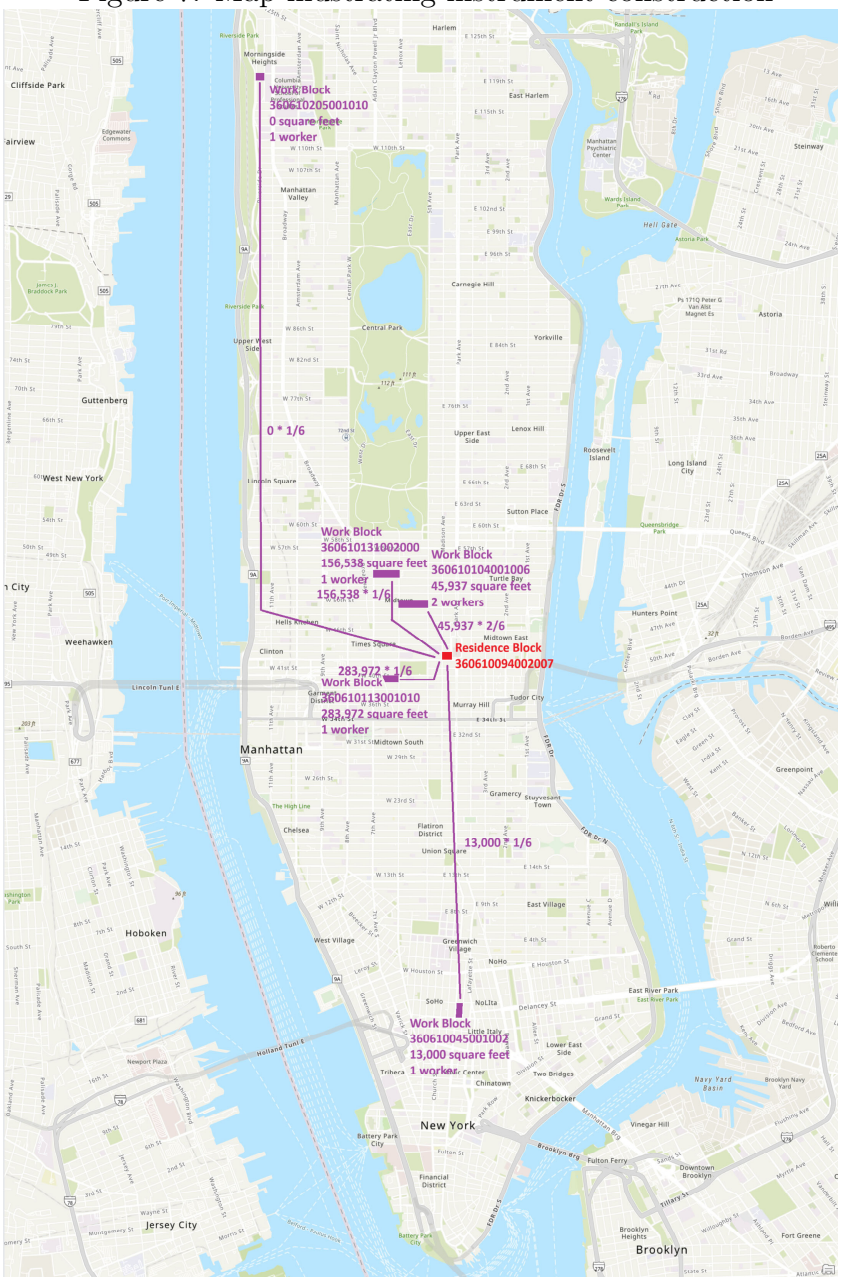
Notes: This figure shows the share of employed individuals working remotely, defined as those reporting "worked at home" as their primary means of transportation to work. It uses ACS data from 2005 to 2023 drawn from the public IPUMS files rather than the restricted-access microdata. It includes all employed ACS respondents, including those in locations without CompStak lease coverage that are not part of the main analysis sample.

Figure 6: Lease termination costs



Notes: This figure illustrates lease termination costs for a stylized 5-year office lease. It assumes a fixed termination cost of 20% of contract value, capturing reputational and lump-sum penalties, and variable costs equal to 50% of the remaining rent. The figure shows that lease adjustment costs fall as expiration approaches but remain significant until expiration, making expiration the key opportunity for firms to adjust leases and adopt remote work.

Figure 7: Map illustrating instrument construction



Notes: This figure illustrates the construction of a Bartik-style measure of exposure to annual office lease expirations. It combines CompStak data on the location and size of expiring office leases with LODES data on employment flows. In this example, the residential block highlighted in red has 6 workers employed in 5 different work blocks highlighted in purple, each associated with a different amount of expiring office space ranging from 0 to 283,972 square feet. The employment shares $\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}$, and $\frac{2}{6}$ are used to compute a weighted average of these expirations, which equals $90,897$ square feet. This weighted average approximates the amount of expiring office space at the workplaces of residents of the red block.

10 Regression Tables

Table 1: First stage

	Remote work (1)	Remote work (2)
Current year expirations	$7.911 \times 10^{-8***}$	
	(1.272×10^{-8})	
Current year expirations X 2005		$-1.722 \times 10^{-7***}$
		(5.471×10^{-8})
Current year expirations X 2006		$-1.939 \times 10^{-7***}$
		(4.062×10^{-8})
Current year expirations X 2007		$-2.436 \times 10^{-7***}$
		(4.192×10^{-8})
Current year expirations X 2008		$-4.297 \times 10^{-7***}$
		(6.155×10^{-8})
Current year expirations X 2009		$-3.939 \times 10^{-7***}$
		(5.824×10^{-8})
Current year expirations X 2010		$-4.370 \times 10^{-7***}$
		(4.751×10^{-8})
Current year expirations X 2011		$-3.250 \times 10^{-7***}$
		(3.12×10^{-8})
Current year expirations X 2012		$-4.609 \times 10^{-7***}$
		(3.838×10^{-8})
Current year expirations X 2013		$-3.603 \times 10^{-7***}$
		(3.423×10^{-8})
Current year expirations X 2014		$-5.146 \times 10^{-7***}$
		(3.529×10^{-8})
Current year expirations X 2015		$-4.626 \times 10^{-7***}$
		(3.107×10^{-8})
Current year expirations X 2016		$-3.581 \times 10^{-7***}$
		(2.639×10^{-8})
Current year expirations X 2017		$-3.598 \times 10^{-7***}$
		(2.469×10^{-8})
Current year expirations X 2018		$-3.120 \times 10^{-7***}$
		(2.287×10^{-8})
Current year expirations X 2019		$-3.855 \times 10^{-7***}$
		(2.547×10^{-8})
Current year expirations X 2020		$6.953 \times 10^{-7***}$
		(4.267×10^{-8})
Current year expirations X 2021		$8.561 \times 10^{-7***}$
		(5.64×10^{-8})
Current year expirations X 2022		$3.301 \times 10^{-7***}$
		(3.725×10^{-8})
Current year expirations X 2023		3.386×10^{-8}
		(2.786×10^{-8})
N	17,440,000	17,440,000
F	38.69	29.72
R^2	0.0739	0.0750
Outcome mean	0.0898	0.0898
Outcome SD	0.2859	0.2859

Notes: This table presents the results of the first stage regression shown in Equation 6. The dependent variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. For instruments, Column (1) uses a Bartik-style measure of current-year office lease expirations defined in Equation 1. Measured in square feet, this approximates the amount of expiring office space at the workplaces of residents in each block during the current year. Column (2) interacts the same measure with calendar year. Both columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: First stage

	Remote work (1)	Remote work (2)
5 year expirations	$1.111 \times 10^{-7***}$	
	(1.148×10^{-8})	
5 year expirations X 2005		$-1.501 \times 10^{-7***}$
		(5.361×10^{-8})
5 year expirations X 2006		$-1.228 \times 10^{-7***}$
		(2.985×10^{-8})
5 year expirations X 2007		$-1.077 \times 10^{-7***}$
		(2.508×10^{-8})
5 year expirations X 2008		$-2.704 \times 10^{-7***}$
		(4.456×10^{-8})
5 year expirations X 2009		$-2.209 \times 10^{-7***}$
		(3.784×10^{-8})
5 year expirations X 2010		$-2.186 \times 10^{-7***}$
		(3.146×10^{-8})
5 year expirations X 2011		$-1.775 \times 10^{-7***}$
		(2.382×10^{-8})
5 year expirations X 2012		$-1.674 \times 10^{-7***}$
		(2.153×10^{-8})
5 year expirations X 2013		$-1.552 \times 10^{-7***}$
		(1.919×10^{-8})
5 year expirations X 2014		$-1.575 \times 10^{-7***}$
		(1.513×10^{-8})
5 year expirations X 2015		$-1.497 \times 10^{-7***}$
		(1.281×10^{-8})
5 year expirations X 2016		$-1.438 \times 10^{-7***}$
		(1.128×10^{-8})
5 year expirations X 2017		$-1.487 \times 10^{-7***}$
		(1.029×10^{-8})
5 year expirations X 2018		$-1.369 \times 10^{-7***}$
		(1.023×10^{-8})
5 year expirations X 2019		$-1.383 \times 10^{-7***}$
		(9.064×10^{-9})
5 year expirations X 2020		$3.026 \times 10^{-7***}$
		(1.763×10^{-8})
5 year expirations X 2021		$4.500 \times 10^{-7***}$
		(2.608×10^{-8})
5 year expirations X 2022		$1.956 \times 10^{-7***}$
		(1.919×10^{-8})
5 year expirations X 2023		$5.999 \times 10^{-8***}$
		(1.426×10^{-8})
N	17,440,000	17,440,000
F	93.67	38.49
R^2	0.0740	0.0761
Outcome mean	0.0898	0.0898
Outcome SD	0.2859	0.2859

Notes: This table presents the results of the first stage regression shown in Equation 6. The dependent variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. For instruments, Column (1) uses a Bartik-style measure of cumulative office lease expirations over the previous five years (t–4 to t), defined in Equation 2. Measured in square feet, this approximates the amount of expiring office space at the workplaces of residents in each block over the previous 5 years. Column (2) interacts the same measure with calendar year. Both columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: First stage

	Remote work (1)	Remote work (2)
Normalized current year expirations	$1.263 \times 10^{-5***}$	
	(4.364×10^{-6})	
Normalized current year expirations X 2005		$-9.243 \times 10^{-5**}$
		(3.908×10^{-5})
Normalized current year expirations X 2006		-2.291×10^{-6}
		(6.147×10^{-6})
Normalized current year expirations X 2007		-1.678×10^{-5}
		(1.379×10^{-5})
Normalized current year expirations X 2008		$-2.461 \times 10^{-4***}$
		(5.109×10^{-5})
Normalized current year expirations X 2009		$-2.695 \times 10^{-4***}$
		(5.485×10^{-5})
Normalized current year expirations X 2010		$-3.395 \times 10^{-4***}$
		(4.548×10^{-5})
Normalized current year expirations X 2011		$-2.388 \times 10^{-4***}$
		(5.783×10^{-5})
Normalized current year expirations X 2012		$-1.617 \times 10^{-4***}$
		(2.708×10^{-5})
Normalized current year expirations X 2013		$-6.641 \times 10^{-5***}$
		(2.39×10^{-5})
Normalized current year expirations X 2014		$-3.203 \times 10^{-4***}$
		(4.272×10^{-5})
Normalized current year expirations X 2015		$-1.897 \times 10^{-4***}$
		(2.171×10^{-5})
Normalized current year expirations X 2016		$-1.331 \times 10^{-4***}$
		(4.008×10^{-5})
Normalized current year expirations X 2017		$-1.347 \times 10^{-4***}$
		(4.194×10^{-5})
Normalized current year expirations X 2018		$-7.966 \times 10^{-5*}$
		(4.069×10^{-5})
Normalized current year expirations X 2019		$-1.788 \times 10^{-4***}$
		(2.51×10^{-5})
Normalized current year expirations X 2020		$3.546 \times 10^{-4***}$
		(4.497×10^{-5})
Normalized current year expirations X 2021		$2.732 \times 10^{-4***}$
		(8.354×10^{-5})
Normalized current year expirations X 2022		$2.203 \times 10^{-4***}$
		(5.627×10^{-5})
Normalized current year expirations X 2023		$4.970 \times 10^{-5***}$
		(1.84×10^{-5})
N	17,440,000	17,440,000
F	8.382	10.81
R^2	0.0739	0.0741
Outcome mean	0.0898	0.0898
Outcome SD	0.2859	0.2859

Notes: This table presents the results of the first stage regression shown in Equation 6. The dependent variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. For instruments, Column (1) uses a Bartik-style measure of current-year office lease expirations defined in Equation 3. Measured in square feet per worker, this approximates the amount of expiring office space at the workplaces of residents in each block during the current year, accounting for number of employees. Column (2) interacts the same measure with calendar year. Both columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: First stage

	Remote work (1)	Remote work (2)
Normalized 5 year expirations	$2.128 \times 10^{-5***}$	
	(2.803×10^{-6})	
Normalized 5 year expirations X 2005		$-9.296 \times 10^{-5**}$
		(3.896×10^{-5})
Normalized 5 year expirations X 2006		-6.173×10^{-6}
		(6.885×10^{-6})
Normalized 5 year expirations X 2007		$-2.316 \times 10^{-5*}$
		(1.362×10^{-5})
Normalized 5 year expirations X 2008		$-1.248 \times 10^{-4***}$
		(3.643×10^{-5})
Normalized 5 year expirations X 2009		$-1.255 \times 10^{-4**}$
		(4.9×10^{-5})
Normalized 5 year expirations X 2010		$-1.236 \times 10^{-4***}$
		(3.171×10^{-5})
Normalized 5 year expirations X 2011		$-1.650 \times 10^{-4***}$
		(2.545×10^{-5})
Normalized 5 year expirations X 2012		$-1.429 \times 10^{-4***}$
		(1.946×10^{-5})
Normalized 5 year expirations X 2013		$-9.266 \times 10^{-5***}$
		(1.501×10^{-5})
Normalized 5 year expirations X 2014		$-7.970 \times 10^{-5***}$
		(1.525×10^{-5})
Normalized 5 year expirations X 2015		$-9.568 \times 10^{-5***}$
		(1.756×10^{-5})
Normalized 5 year expirations X 2016		$-5.040 \times 10^{-5***}$
		(1.225×10^{-5})
Normalized 5 year expirations X 2017		$-9.234 \times 10^{-5***}$
		(1.45×10^{-5})
Normalized 5 year expirations X 2018		$-9.483 \times 10^{-5***}$
		(1.706×10^{-5})
Normalized 5 year expirations X 2019		$-1.305 \times 10^{-4***}$
		(1.247×10^{-5})
Normalized 5 year expirations X 2020		$2.200 \times 10^{-4***}$
		(2.21×10^{-5})
Normalized 5 year expirations X 2021		$2.557 \times 10^{-4***}$
		(3.658×10^{-5})
Normalized 5 year expirations X 2022		$1.626 \times 10^{-4***}$
		(2.269×10^{-5})
Normalized 5 year expirations X 2023		$4.309 \times 10^{-5***}$
		(9.597×10^{-6})
N	17,440,000	17,440,000
F	57.65	14.55
R^2	0.0739	0.0746
Outcome mean	0.0898	0.0898
Outcome SD	0.2859	0.2859

Notes: This table presents the results of the first stage regression shown in Equation 6. The dependent variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. For instruments, Column (1) uses a Bartik-style measure of cumulative office lease expirations over the previous five years (t–4 to t), defined in Equation 4. Measured in square feet per worker, this approximates the amount of expiring office space per worker at the workplaces of residents in each block over the previous 5 years, accounting for employment size. Column (2) interacts the same measure with calendar year. Both columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Effects of remote work on tax sorting

	Migrate	State income tax	City income tax	Property tax
	(1)	(2)	(3)	(4)
Remote work	0.1132***	-0.01997***	-0.04047***	431.1***
	(0.01092)	(0.002647)	(0.006307)	(143.3)
N	17,440,000	17,440,000	17,440,000	17,440,000
R^2	0.0000	-0.0050	-0.0240	0.1960
Outcome mean	0.0961	0.8311	0.0705	2959
Outcome SD	0.2947	0.3747	0.2560	4082

Notes: This table presents the results of the second stage regression shown in Equation 5, which estimates the causal impact of remote work on selected outcomes. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. The first outcome indicates changing residences in the past year. The next 2 outcomes are indicators that the worker lives in a state or city that levies income taxes. The final outcome is annual property taxes paid by the worker's household, set to 0 for renters. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Effects of remote work on school utilization

	Private school	Public school	Migrate × private school	Migrate × public school
	(1)	(2)	(3)	(4)
Remote work	-0.1142***	0.06526***	0.001386	0.02813***
	(0.008549)	(0.0084)	(0.002406)	(0.004301)
N	17,440,000	17,440,000	17,440,000	17,440,000
R^2	-0.0150	-0.0050	0.0000	-0.0010
Outcome mean	0.0654	0.2929	0.0047	0.0209
Outcome SD	0.2472	0.4551	0.0685	0.1430

Notes: This table presents the results of the second stage regression shown in Equation 5, which estimates the causal impact of remote work on selected outcomes. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. The first 2 outcomes indicate whether the worker's household has a child enrolled in private or public school, proxying for parental school choice. The final 2 outcomes interact these enrollment indicators with migration as defined in Table 5. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Housing

	Cost	Value	Ownership	House	Bedrooms	Rooms	Large lot
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Remote Work	130.2***	43,270*	0.07688***	0.07748***	0.1552***	0.3593***	0.04361***
	(41.22)	(24,390)	(0.01608)	(0.01428)	(0.04292)	(0.09698)	(0.01076)
N	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000
R^2	0.1510	0.1670	0.1420	0.1820	0.1260	0.1290	0.1240
Outcome mean	1,643	258,000	0.7325	0.7140	3.051	6.460	0.1405
Outcome SD	1,203	465,000	0.4427	0.4519	1.133	2.482	0.3475

Notes: This table presents the results of the second stage regression shown in Equation 5, which estimates the causal impact of remote work on selected housing outcomes. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. As in previous tables, each observation is an individual worker, but all housing outcomes are defined at the worker's household level and thus are shared across workers in the same household. The first 3 outcomes in Columns (1), (2), and (3) measure housing finance, including monthly gross housing expenditures, self-reported home value (renters coded as 0), and an ownership indicator. The last 4 outcomes capture housing characteristics, starting with an indicator for living in a detached single family home rather than a townhouse, multifamily unit, or mobile home. Interior living space demand is measured in Columns (5) and (6) by the number of rooms and bedrooms. Exterior space demand is measured in Column (7) by an indicator for residing on a lot of at least 1 acre, which reflects ACS variable construction. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Effects of remote work on neighborhood choice

	Child Sorting (1)	Racial Sorting (2)	Income Relative (3)	Income Relative^2 (4)	Neighborhood Income $\Delta\Delta$ (5)	Neighborhood Income $(\Delta\Delta)^2$ (6)	Neighborhood Income $\lvert\Delta\Delta\rvert$ (7)
Remote work	-0.02188**	-0.02621*	-3,410	12,210,000,000***	42.63	23,330,000	1,847***
	(0.009657)	(0.01565)	(4,110)	(2,493,000,000)	(439.6)	(89,460,000)	(311.4)
N	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000
R^2	0.0120	0.0070	0.0094	0.0170	0.0580	0.0040	-0.0050
Outcome mean	0.2116	0.7543	21600	5170000000	6.249	-7,000,000	-28.41
Outcome SD	0.4084	0.4305	68600	40100000000	7,281	1,600,000,000	6,475

Notes: This table presents the results of the second stage regression shown in Equation 5, which estimates the causal impact of remote work on selected neighborhood sorting outcomes. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. The first 2 outcomes measure sorting based on whether workers live in neighborhoods that match their child status or racial group, the next 2 capture the linear and squared difference between individual and neighborhood income, and the final 3 report difference-in-differences measures of neighborhood income in linear, squared, and absolute value form. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Placebo test - public finance outcomes

	State income tax	Local income tax	Property tax	Public school	Private school
	(1)	(2)	(3)	(4)	(5)
Remote work	0.004773*	-0.003257	27.32	0.02545***	-0.002717
	(0.002861)	(0.002760)	(72.21)	(0.006279)	(0.002198)
N	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000
F	2.782	1.393	0.1431	16.43	1.528
Placebo ratio $\frac{\beta_{main}}{\beta_{placebo}}$	4.184	12.42	15.78	2.564	42.03
Outcome mean	0.8311	0.0705	2959	0.2929	0.0654
Outcome SD	0.0516	0.0497	1427	0.1914	0.0511

Notes: This table presents placebo tests for public finance outcomes. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. Each placebo outcome is constructed by regressing selected tax and school enrollment outcomes from Tables 5 and 6 on a set of demographic controls and constructing fitted values, as described in Section 7.1. These placebo outcomes are then used in the second stage regression shown in Equation 5. The row "Placebo ratio" reports the ratio of each coefficient in Tables 5 and 6 to its corresponding placebo coefficient here. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Placebo test - residential choice outcomes

	Migrate	Cost	House	Bedrooms	Income $\lvert\Delta\Delta\rvert$
	(1)	(2)	(3)	(4)	(5)
Remote work	-0.01900***	-11.48	0.01053*	0.01325	11.24***
	(0.003175)	(22.64)	(0.006004)	(0.01211)	(4.094)
N	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000
F	35.82	0.2573	3.076	1.198	7.532
Placebo ratio $\frac{\beta_{main}}{\beta_{placebo}}$	5.958	11.34	7.358	11.71	164.3
Outcome mean	0.0961	1643	0.7140	3.051	-28.41
Outcome SD	0.0704	416.8	0.1259	0.2492	93.33

Notes: This table presents placebo tests for residential choice outcomes. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. Each outcome is constructed by regressing selected migration, housing, and neighborhood choice outcomes from Tables 5, 7, and 8 on a set of demographic controls and constructing fitted values, as described in Section 7.1. These placebo outcomes are then used in the second stage regression shown in Equation 5. The row "Placebo ratio" reports the ratio of each coefficient in Tables 5, 7, and 8 to its corresponding placebo coefficient here. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: Robustness to instrument type - migration outcome

	Migrate (1)	Migrate (2)	Migrate (3)	Migrate (4)	Migrate (5)	Migrate (6)	Migrate (7)	Migrate (8)
Remote work	0.2516***	0.1631***	0.3292***	0.1612***	0.5288*	0.1402***	0.1302*	0.1132***
	(0.0805)	(0.01254)	(0.04499)	(0.01189)	(0.3204)	(0.01926)	(0.07528)	(0.01092)
N	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000
R^2	-0.0446	-0.0119	-0.0849	-0.0113	-0.2387	-0.0057	-0.0033	0.0003
Outcome mean	0.0961	0.0961	0.0961	0.0961	0.0961	0.0961	0.0961	0.0961
Outcome SD	0.2947	0.2947	0.2947	0.2947	0.2947	0.2947	0.2947	0.2947

Notes: This table presents robustness to varying instrument variable functional forms. The outcome is an indicator for migration in the past year. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. Columns (1) and (2) use the annual square-footage instrument from Equation 1, Columns (3) and (4) use the five-year cumulative square-footage instrument from Equation 2, Columns (5) and (6) use the annual square-footage-per-worker instrument from Equation 3, and Columns (7) and (8) use the five-year cumulative square-footage-per-worker instrument from Equation 4. Within each pair, the even-numbered column interacts the instrument with calendar year; Column (8) is the baseline specification used in Tables 5, 6, 7, and 8. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 12: Robustness to instrument type - public school enrollment outcome

	Public school (1)	Public school (2)	Public school (3)	Public school (4)	Public school (5)	Public school (6)	Public school (7)	Public school (8)
Remote Work	0.1819***	0.06491***	0.1278***	0.06816***	0.3222	0.07868***	0.1916***	0.06526***
	(0.06228)	(0.007475)	(0.0206)	(0.006085)	(0.1965)	(0.01288)	(0.06171)	(0.0084)
N	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000
R^2	-0.0522	-0.0048	-0.025	-0.0056	-0.165	-0.0082	-0.058	-0.0049
Outcome mean	0.2929	0.2929	0.2929	0.2929	0.2929	0.2929	0.2929	0.2929
Outcome SD	0.4551	0.4551	0.4551	0.4551	0.4551	0.4551	0.4551	0.4551

Notes: This table presents robustness to varying instrument variable functional forms. The outcome is an indicator for the presence of children in the household enrolled in public school, which proxies for school enrollment choices of parents. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. Columns (1) and (2) use the annual square-footage instrument from Equation 1, Columns (3) and (4) use the five-year cumulative square-footage instrument from Equation 2, Columns (5) and (6) use the annual square-footage-per-worker instrument from Equation 3, and Columns (7) and (8) use the five-year cumulative square-footage-per-worker instrument from Equation 4. Within each pair, the even-numbered column interacts the instrument with calendar year; Column (8) is the baseline specification used in Tables 5, 6, 7, and 8. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 13: Robustness to controls - migration outcome

	Migrate (1)	Migrate (2)	Migrate (3)	Migrate (4)
Remote work	0.1133***	0.1698***	0.1310***	0.1132***
	(0.01091)	(0.01164)	(0.01117)	(0.01949)
Expected expirations	-0.00000007322			
	(0.0000001978)			
N	17,440,000	17,440,000	17,440,000	17,440,000
R^2	0.0003	-0.0236	-0.0059	0.0003
Outcome mean	0.0961	0.0961	0.0961	0.0961
Outcome SD	0.2947	0.2947	0.2947	0.2947

Notes: This table presents robustness to varying controls and clustering. The outcome is an indicator for migration in the past year. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. Column (1), following Borusyak et al. (2025), adds a control for the expected value of the instrument. Column (2) adds fixed effects for detailed Census industry, and Column (3) includes these plus a full set of demographic controls listed in Equation 11. Columns (1), (2), and (3) cluster by interactions of three-digit NAICS industry, city, and county. Column (4) omits the additional controls in (1), (2), and (3), and instead uses multiway clustering. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

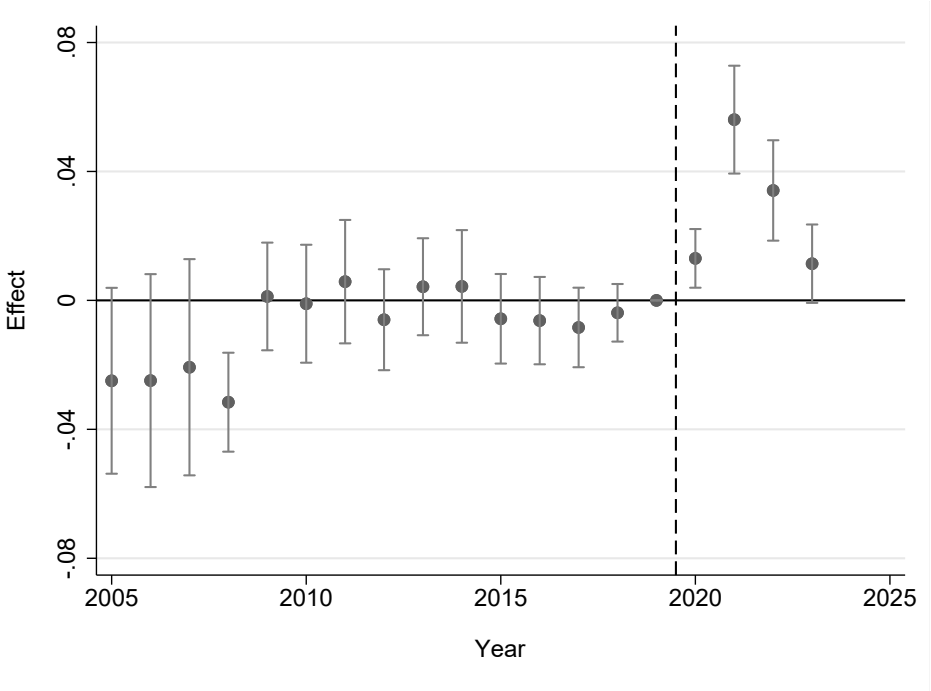
Table 14: Robustness to controls - public school enrollment outcome

	Public school (1)	Public school (2)	Public school (3)	Public school (4)
Remote Work	0.06534***	0.03536***	0.06125***	0.06526***
	(0.00841)	(0.007378)	(0.008131)	(0.01565)
Expected expirations	-0.0000002528*			
	(0.0000001445)			
N	17,440,000	17,440,000	17,440,000	17,440,000
R^2	-0.0049	-0.003	-0.005	-0.0049
Outcome mean	0.2929	0.2929	0.2929	0.2929
Outcome SD	0.4551	0.4551	0.4551	0.4551

Notes: This table presents robustness to varying controls and clustering. The outcome is an indicator for the presence of children in the household enrolled in public school, which proxies for school enrollment choices of parents. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. Column (1), following Borusyak et al. (2025), adds a control for the expected value of the instrument. Column (2) adds fixed effects for detailed Census industry, and Column (3) includes these plus a full set of demographic controls listed in Equation 11. Columns (1), (2), and (3) cluster by interactions of three-digit NAICS industry, city, and county. Column (4) omits the additional controls in (1), (2), and (3), and instead uses multiway clustering. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

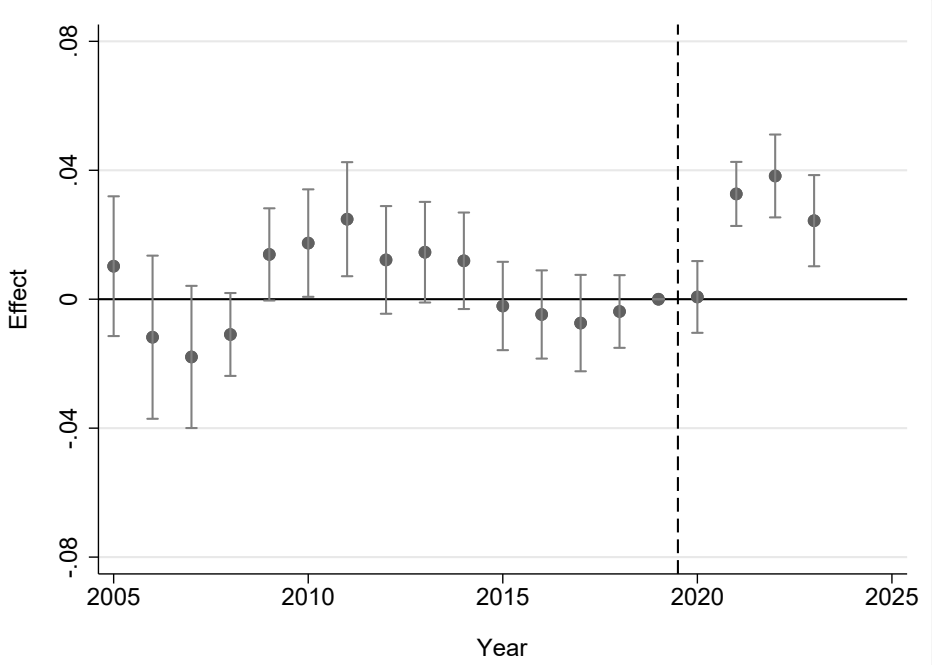
11 Event Studies

Figure 8: Neighborhood outward migration



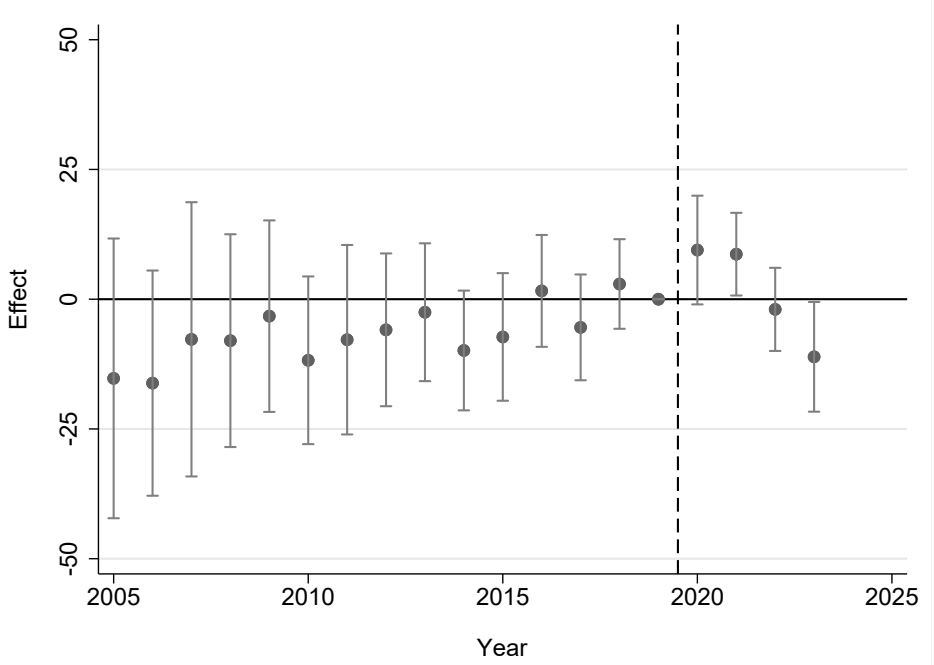
Notes: This figure shows event study coefficients and 95% confidence intervals from the instrumented dynamic difference-in-differences model shown in Equation 9. The treatment group is neighborhoods (Census tracts) with above-average remote work shares among residents in 2020, instrumented using tract-level office lease expirations as described in Equation 10. Each point gives the effect of high remote work on the neighborhood's annual outward migration rate in a given event year, relative to the omitted base year 2019. The vertical dashed line designates neighborhoods surveyed in 2020 and onward as the treated observations, reflecting the sudden rise in remote work that year. The regression includes fixed effects for year and tract, and controls for local office real estate characteristics. All standard errors are clustered by county.

Figure 9: Neighborhood inward migration



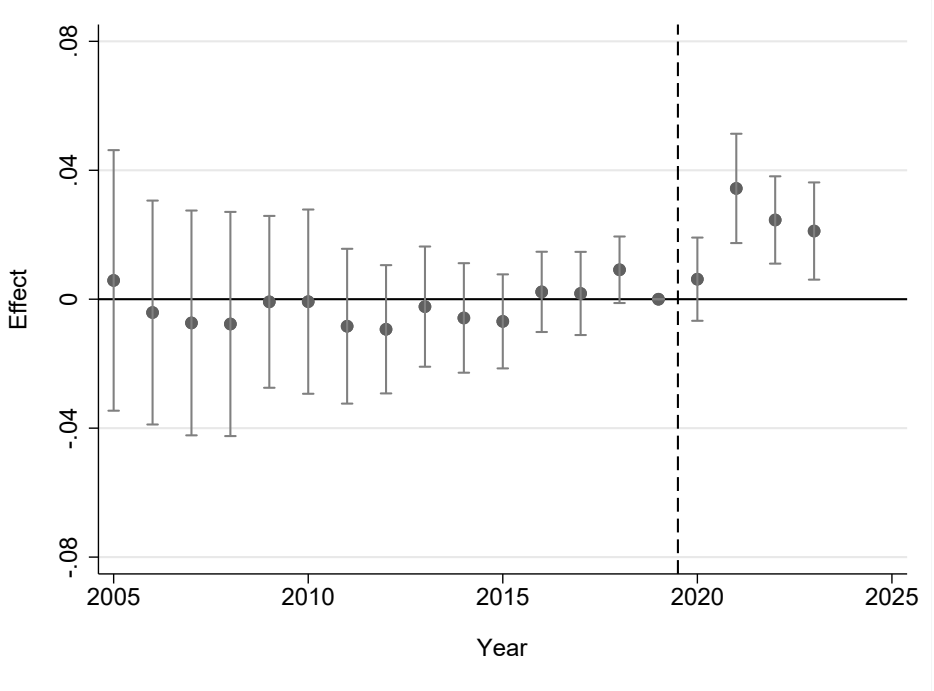
Notes: This figure shows event study coefficients and 95% confidence intervals from the instrumented dynamic difference-in-differences model shown in Equation 9. The treatment group is neighborhoods (Census tracts) with above-average remote work shares among residents in 2020, instrumented using tract-level office lease expirations as described in Equation 10. Each point gives the effect of high remote work on the neighborhood's annual inward migration rate in a given event year, relative to the omitted base year 2019. The vertical dashed line designates neighborhoods surveyed in 2020 and onward as the treated observations, reflecting the sudden rise in remote work that year. The regression includes fixed effects for year and tract, and controls for local office real estate characteristics. All standard errors are clustered by county.

Figure 10: Neighborhood variance in age



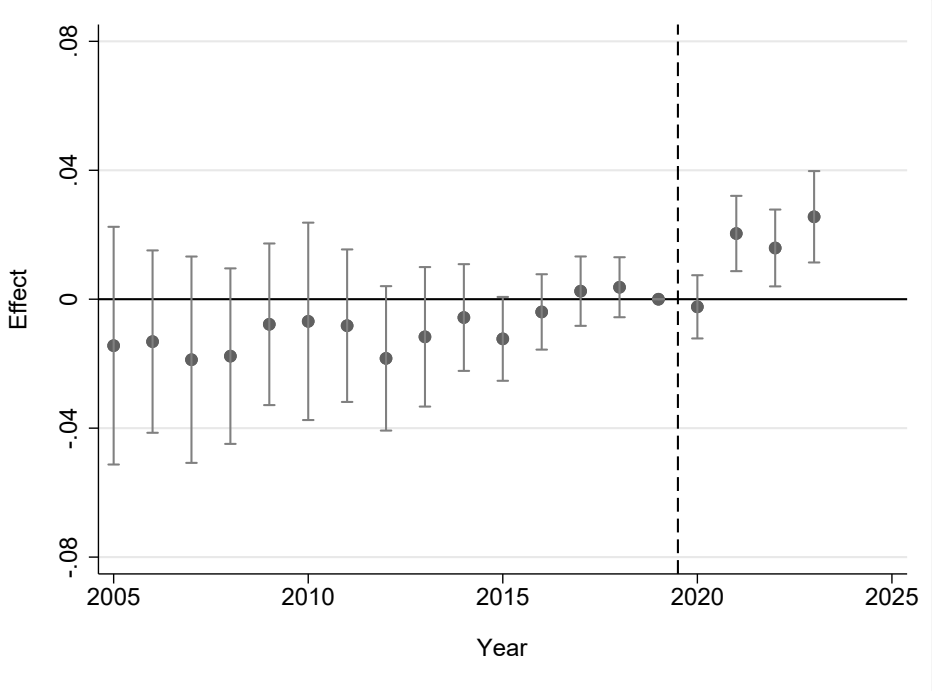
Notes: This figure shows event study coefficients and 95% confidence intervals from the instrumented dynamic difference-in-differences model shown in Equation 9. The treatment group is neighborhoods (Census tracts) with above-average remote work shares among residents in 2020, instrumented using tract-level office lease expirations as described in Equation 10. Each point gives the effect of high remote work on sorting by age, measured by the variance among residents in a given event year, relative to the omitted base year 2019. The vertical dashed line designates neighborhoods surveyed in 2020 and onward as the treated observations, reflecting the sudden rise in remote work that year. The regression includes fixed effects for year and tract, and controls for local office real estate characteristics. All standard errors are clustered by county.

Figure 11: Neighborhood Gini coefficient of income



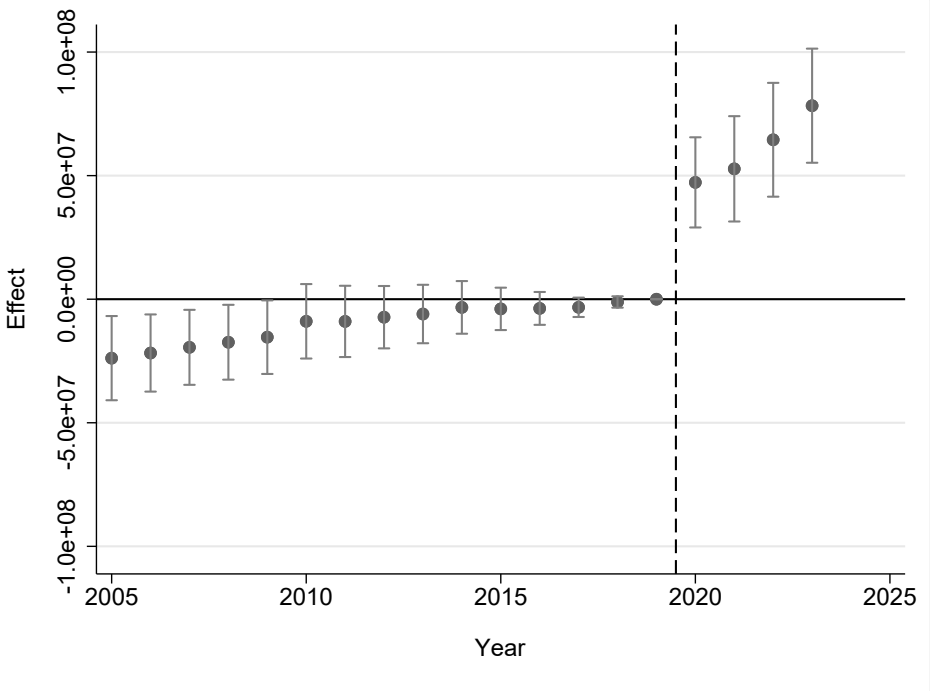
Notes: This figure shows event study coefficients and 95% confidence intervals from the instrumented dynamic difference-in-differences model shown in Equation 9. The treatment group is neighborhoods (Census tracts) with above-average remote work shares among residents in 2020, instrumented using tract-level office lease expirations as described in Equation 10. Each point gives the effect of high remote work on income sorting, measured by the neighborhood's Gini coefficient of income among residents in a given event year, relative to the omitted base year 2019. The vertical dashed line designates neighborhoods surveyed in 2020 and onward as the treated observations, reflecting the sudden rise in remote work that year. The regression includes fixed effects for year and tract, and controls for local office real estate characteristics. All standard errors are clustered by county.

Figure 12: Neighborhood HHI in school enrollment



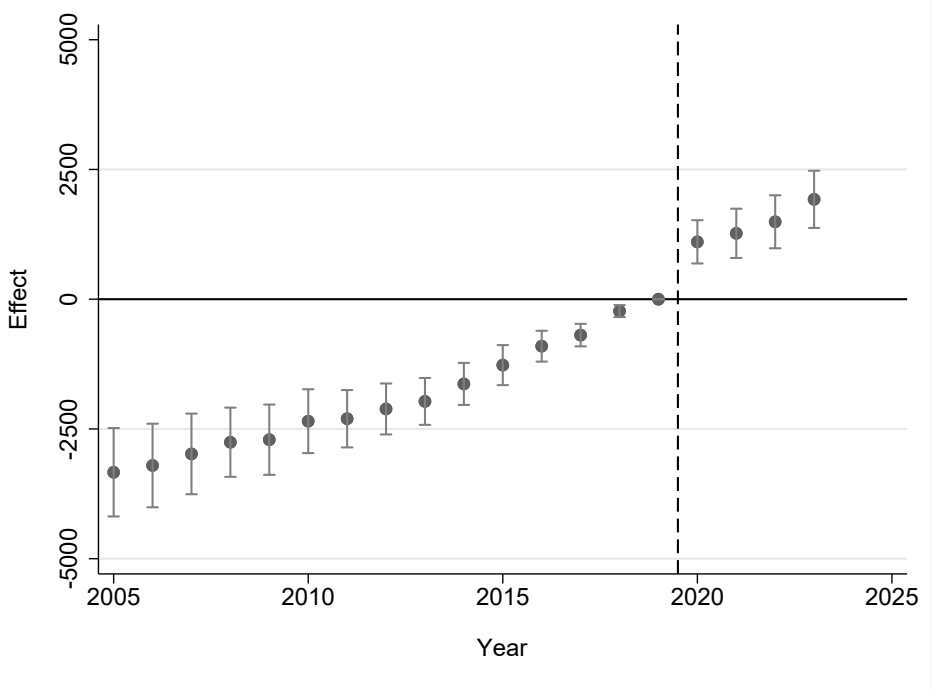
Notes: This figure shows event study coefficients and 95% confidence intervals from the instrumented dynamic difference-in-differences model shown in Equation 9. The treatment group is neighborhoods (Census tracts) with above-average remote work shares among residents in 2020, instrumented using tract-level office lease expirations as described in Equation 10. Each point gives the effect of high remote work on sorting by school enrollment type (private, public, none), measured by the Herfindahl-Hirschman index among households in a given event year, relative to the omitted base year 2019. The vertical dashed line designates neighborhoods surveyed in 2020 and onward as the treated observations, reflecting the sudden rise in remote work that year. The regression includes fixed effects for year and tract, and controls for local office real estate characteristics. All standard errors are clustered by county.

Figure 13: Neighborhood variance in property taxes



Notes: This figure shows event study coefficients and 95% confidence intervals from the instrumented dynamic difference-in-differences model shown in Equation 9. The treatment group is neighborhoods (Census tracts) with above-average remote work shares among residents in 2020, instrumented using tract-level office lease expirations as described in Equation 10. Each point gives the effect of high remote work on variation in property taxes paid by households, relative to the omitted base year 2019. The vertical dashed line designates neighborhoods surveyed in 2020 and onward as the treated observations, reflecting the sudden rise in remote work that year. The regression includes fixed effects for year and tract, and controls for local office real estate characteristics. All standard errors are clustered by county.

Figure 14: Neighborhood average property taxes



Notes: This figure shows event study coefficients and 95% confidence intervals from the instrumented dynamic difference-in-differences model shown in Equation 9. The treatment group is neighborhoods (Census tracts) with above-average remote work shares among residents in 2020, instrumented using tract-level office lease expirations as described in Equation 10. Each point gives the effect of high remote work on property taxes paid per household in a given event year, relative to the omitted base year 2019. The vertical dashed line designates neighborhoods surveyed in 2020 and onward as the treated observations, reflecting the sudden rise in remote work that year. The regression includes fixed effects for year and tract, and controls for local office real estate characteristics. All standard errors are clustered by county.

ONLINE APPENDICES

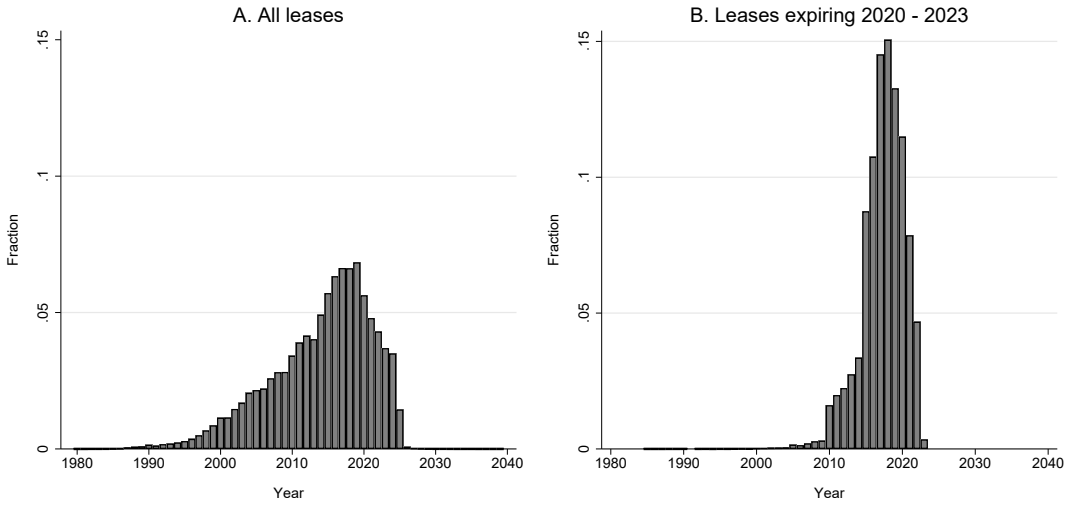
Remote Work and Residential Sorting:
IV Evidence From Expiring Office Leases

Richard De Thorpe

A Appendix

A.1 Figures

Figure A1: Distribution of office lease contract start years



Notes: This figure shows the time coverage of the CompStak office-lease dataset. Panel A shows the number of lease contracts by lease start year, with both coverage and newly contracted office space peaking in 2019. A small forward-dated tail reflects leases scheduled to begin after 2024. Panel B restricts to leases expiring in 2020–2023, the contracts that provide the strongest quasi-random variation in remote work. Because large office leases have longer terms, 90% of this square footage comes from contracts signed in 2019 or earlier.

A.2 Tables

Table A1: Migration and jurisdictional choice

	Exit state with income tax (1)	Enter state with income tax (2)	Exit city with income tax (3)	Enter city with income tax (4)
Remote work	0.01264***	-0.007334***	0.03923***	-0.001235
	(0.002372)	(0.0009662)	(0.005882)	(0.002174)
N	17,440,000	17,440,000	17,440,000	17,440,000
R^2	-0.0020	-0.0020	-0.0370	0.0000
Outcome mean	0.0030	0.0025	0.003	0.0021
Outcome SD	0.0545	0.0495	0.0546	0.0454

Notes: This table presents the results of the second stage regression shown in Equation 5, which estimates the causal impact of remote work on selected outcomes. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. To supplement the outcomes in Table 5, this table has alternative measures of income tax sorting, at the state level in Columns (1) and (2), and at the city level in Columns (3) and (4). Columns (1) and (3) equal 1 if an individual moved from a jurisdiction with an income tax to one without an income tax, while Columns (2) and (4) equal 1 if an individual moved from a jurisdiction without an income tax to one with an income tax. All variables equal 0 if the individual did not move or did not change income tax regime through migration. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Robustness to instrument type - state tax outcome

	State tax	State tax	State tax	State tax	State tax	State tax	State tax	State tax
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Remote Work	-0.06427***	-0.02021***	-0.04878***	-0.02306***	-0.09481	-0.02637***	-0.01986*	-0.01997***
	(0.0195)	(0.002689)	(0.008283)	(0.002621)	(0.06109)	(0.004424)	(0.01161)	(0.002647)
N	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000
R^2	-0.0562	-0.0051	-0.0320	-0.0067	-0.1239	-0.0089	-0.0049	-0.0050
Outcome mean	0.8311	0.8311	0.8311	0.8311	0.8311	0.8311	0.8311	0.8311
Outcome SD	0.3747	0.3747	0.3747	0.3747	0.3747	0.3747	0.3747	0.3747

Notes: This table presents robustness to varying instrument variable functional forms. The outcome is an indicator for living in a state that levies income taxes. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. Columns (1) and (2) use the annual square-footage instrument from Equation 1, Columns (3) and (4) use the five-year cumulative square-footage instrument from Equation 2, Columns (5) and (6) use the annual square-footage-per-worker instrument from Equation 3, and Columns (7) and (8) use the five-year cumulative square-footage-per-worker instrument from Equation 4. Within each pair, the even-numbered column interacts the instrument with calendar year; Column (8) is the baseline specification used in Tables 5, 6, 7, and 8. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Robustness to instrument type - city tax outcome

	City tax (1)	City tax (2)	City tax (3)	City tax (4)	City tax (5)	City tax (6)	City tax (7)	City tax (8)
Remote Work	-0.1457***	-0.05094***	-0.1525***	-0.05380***	0.01267	-0.04920***	-0.09958***	-0.04047***
	(0.0328)	(0.005444)	(0.02229)	(0.005665)	(0.07703)	(0.008462)	(0.03842)	(0.006307)
N	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000	17,440,000
R^2	-0.3260	-0.0390	-0.3580	-0.0430	-0.0020	-0.0360	-0.1510	-0.0240
Outcome mean	0.0705	0.0705	0.0705	0.0705	0.0705	0.0705	0.0705	0.0705
Outcome SD	0.2560	0.2560	0.2560	0.2560	0.2560	0.2560	0.2560	0.2560

Notes: This table presents robustness to varying instrument variable functional forms. The outcome is an indicator for living in a city that levies income taxes. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. Columns (1) and (2) use the annual square-footage instrument from Equation 1, Columns (3) and (4) use the five-year cumulative square-footage instrument from Equation 2, Columns (5) and (6) use the annual square-footage-per-worker instrument from Equation 3, and Columns (7) and (8) use the five-year cumulative square-footage-per-worker instrument from Equation 4. Within each pair, the even-numbered column interacts the instrument with calendar year; Column (8) is the baseline specification used in Tables 5, 6, 7, and 8. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All standard errors are clustered by interactions of three-digit NAICS industry, city, and county. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Robustness to controls - state tax outcome

	State tax	State tax	State tax	State tax
	(1)	(2)	(3)	(4)
Remote Work	-0.01999***	-0.02018***	-0.02020***	-0.01997***
	(0.002646)	(0.002634)	(0.002624)	(0.005367)
Expected expirations	0.00000004352			
	(0.00000004144)			
N	17,440,000	17,440,000	17,440,000	17,440,000
R^2	-0.0050	-0.0051	-0.0047	-0.0050
Outcome mean	0.8311	0.8311	0.8311	0.8311
Outcome SD	0.3747	0.3747	0.3747	0.3747

Notes: This table presents robustness to varying controls and clustering. The outcome is an indicator for living in a state that levies income taxes. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. Column (1), following Borusyak et al. (2025), adds a control for the expected value of the instrument. Column (2) adds fixed effects for detailed Census industry, and Column (3) includes these plus a full set of demographic controls listed in Equation 11. Columns (1), (2), and (3) cluster by interactions of three-digit NAICS industry, city, and county. Column (4) omits the additional controls in (1), (2), and (3), and instead uses multiway clustering. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Robustness to controls - city tax outcome

	City tax (1)	City tax (2)	City tax (3)	City tax (4)
Remote Work	-0.04055***	-0.04185***	-0.04009***	-0.04047***
	(0.006308)	(0.006193)	(0.006244)	(0.008689)
Expected expirations	0.0000001852***			
	(0.00000005971)			
N	17,440,000	17,440,000	17,440,000	17,440,000
R^2	-0.0240	-0.0260	-0.0220	-0.0240
Outcome mean	0.0705	0.0705	0.0705	0.0705
Outcome SD	0.2560	0.2560	0.2560	0.2560

Notes: This table presents robustness to varying controls and clustering. The outcome is an indicator for living in a city that levies income taxes. The explanatory variable is $\mathbf{1}\{remote\}$, an indicator for self-reported remote work, defined for employed individuals only. Column (1), following Borusyak et al. (2025), adds a control for the expected value of the instrument. Column (2) adds fixed effects for detailed Census industry, and Column (3) includes these plus a full set of demographic controls listed in Equation 11. Columns (1), (2), and (3) cluster by interactions of three-digit NAICS industry, city, and county. Column (4) omits the additional controls in (1), (2), and (3), and instead uses multiway clustering. All columns include controls for local office real estate characteristics and year fixed effects. To condition on prior year place of residence, they include linear controls for neighborhood income and housing characteristics, as well as fixed effects for the prior year city and county of residence, as described in Section 3.3. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Robustness to controls - first stage

	Remote work (1)	Remote work (2)	Remote work (3)	Remote work (4)
5 year expirations/worker × 2005	$-9.256 \times 10^{-5**}$	$-1.314 \times 10^{-4***}$	$-1.159 \times 10^{-4***}$	$-9.296 \times 10^{-5**}$
	(3.895×10^{-5})	(3.856×10^{-5})	(4.232×10^{-5})	(4.215×10^{-5})
5 year expirations/worker × 2006	-6.308×10^{-6}	-5.407×10^{-6}	-3.443×10^{-6}	-6.173×10^{-6}
	(6.897×10^{-6})	(6.795×10^{-6})	(7.004×10^{-6})	(1.044×10^{-5})
5 year expirations/worker × 2007	$-2.374 \times 10^{-5*}$	-7.131×10^{-6}	$-2.271 \times 10^{-5*}$	-2.316×10^{-5}
	(1.367×10^{-5})	(1.266×10^{-5})	(1.272×10^{-5})	(1.695×10^{-5})
5 year expirations/worker × 2008	$-1.245 \times 10^{-4***}$	$-1.490 \times 10^{-4***}$	$-1.350 \times 10^{-4***}$	$-1.248 \times 10^{-4**}$
	(3.639×10^{-5})	(4.116×10^{-5})	(3.434×10^{-5})	(5.405×10^{-5})
5 year expirations/worker × 2009	$-1.256 \times 10^{-4**}$	$-1.592 \times 10^{-4***}$	$-1.366 \times 10^{-4***}$	$-1.255 \times 10^{-4**}$
	(4.901×10^{-5})	(5.59×10^{-5})	(4.712×10^{-5})	(5.807×10^{-5})
5 year expirations/worker × 2010	$-1.238 \times 10^{-4***}$	$-1.528 \times 10^{-4***}$	$-1.212 \times 10^{-4***}$	$-1.236 \times 10^{-4***}$
	(3.173×10^{-5})	(3.829×10^{-5})	(3.065×10^{-5})	(4.645×10^{-5})
5 year expirations/worker × 2011	$-1.652 \times 10^{-4***}$	$-2.032 \times 10^{-4***}$	$-1.728 \times 10^{-4***}$	$-1.650 \times 10^{-4***}$
	(2.546×10^{-5})	(2.827×10^{-5})	(2.437×10^{-5})	(4.04×10^{-5})
5 year expirations/worker × 2012	$-1.433 \times 10^{-4***}$	$-1.715 \times 10^{-4***}$	$-1.445 \times 10^{-4***}$	$-1.429 \times 10^{-4***}$
	(1.944×10^{-5})	(2.147×10^{-5})	(1.809×10^{-5})	(3.669×10^{-5})
5 year expirations/worker × 2013	$-9.295 \times 10^{-5***}$	$-1.076 \times 10^{-4***}$	$-9.517 \times 10^{-5***}$	$-9.266 \times 10^{-5***}$
	(1.502×10^{-5})	(1.633×10^{-5})	(1.479×10^{-5})	(2.691×10^{-5})
5 year expirations/worker × 2014	$-8.016 \times 10^{-5***}$	$-8.744 \times 10^{-5***}$	$-8.337 \times 10^{-5***}$	$-7.970 \times 10^{-5**}$
	(1.526×10^{-5})	(1.66×10^{-5})	(1.586×10^{-5})	(3.838×10^{-5})
5 year expirations/worker × 2015	$-9.626 \times 10^{-5***}$	$-9.910 \times 10^{-5***}$	$-9.713 \times 10^{-5***}$	$-9.568 \times 10^{-5***}$
	(1.766×10^{-5})	(1.809×10^{-5})	(1.811×10^{-5})	(2.487×10^{-5})
5 year expirations/worker × 2016	$-5.086 \times 10^{-5***}$	$-4.842 \times 10^{-5***}$	$-5.357 \times 10^{-5***}$	$-5.040 \times 10^{-5**}$
	(1.228×10^{-5})	(1.177×10^{-5})	(1.302×10^{-5})	(2.221×10^{-5})
5 year expirations/worker × 2017	$-9.281 \times 10^{-5***}$	$-8.331 \times 10^{-5***}$	$-9.720 \times 10^{-5***}$	$-9.234 \times 10^{-5***}$
	(1.456×10^{-5})	(1.335×10^{-5})	(1.50×10^{-5})	(2.535×10^{-5})
5 year expirations/worker × 2018	$-9.581 \times 10^{-5***}$	$-8.197 \times 10^{-5***}$	$-1.014 \times 10^{-4***}$	$-9.483 \times 10^{-5***}$
	(1.691×10^{-5})	(1.477×10^{-5})	(1.816×10^{-5})	(2.658×10^{-5})
5 year expirations/worker × 2019	$-1.310 \times 10^{-4***}$	$-1.056 \times 10^{-4***}$	$-1.361 \times 10^{-4***}$	$-1.305 \times 10^{-4***}$
	(1.251×10^{-5})	(1.04×10^{-5})	(1.339×10^{-5})	(2.616×10^{-5})
5 year expirations/worker × 2020	$2.194 \times 10^{-4***}$	$2.498 \times 10^{-4***}$	$2.111 \times 10^{-4***}$	$2.200 \times 10^{-4***}$
	(2.207×10^{-5})	(2.457×10^{-5})	(2.091×10^{-5})	(3.716×10^{-5})
5 year expirations/worker × 2021	$2.552 \times 10^{-4***}$	$2.772 \times 10^{-4***}$	$2.510 \times 10^{-4***}$	$2.557 \times 10^{-4***}$
	(3.654×10^{-5})	(3.915×10^{-5})	(3.569×10^{-5})	(7.925×10^{-5})
5 year expirations/worker × 2022	$1.620 \times 10^{-4***}$	$2.022 \times 10^{-4***}$	$1.581 \times 10^{-4***}$	$1.626 \times 10^{-4***}$
	(2.262×10^{-5})	(2.686×10^{-5})	(2.157×10^{-5})	(3.973×10^{-5})
5 year expirations/worker × 2023	$4.251 \times 10^{-5***}$	$8.487 \times 10^{-5***}$	$3.811 \times 10^{-5***}$	$4.309 \times 10^{-5**}$
	(9.553×10^{-6})	(1.184×10^{-5})	(8.928×10^{-6})	(1.94×10^{-5})
Expected expirations	$1.018 \times 10^{-6***}$			
	(2.268×10^{-7})			
N	17,440,000	17,440,000	17,440,000	17,440,000
F	14.60	14.40	14.45	6.068
R^2	0.0746	0.0693	0.1419	0.0746
Outcome mean	0.0898	0.0898	0.0898	0.0898
Outcome SD	0.2859	0.2859	0.2859	0.2859

Notes: This table presents the robustness of the first stage regression corresponding to the robustness tests in Tables 13 and 14 with varying control sets and clustering. All columns use the Bartik-style five-year cumulative lease-expiration instrument from Equation 4, measured in square feet per worker and interacted with calendar year. Column (1) includes a control for the expected value of the instrument. Column (2) adds fixed effects for detailed Census industry, and Column (3) further adds a full set of demographic controls listed in Equation 11, and Column (4) replaces interactive clustering by city, county, and three-digit NAICS industry in Columns (1), (2), and (3) with multiway clustering. All figures are rounded to 4 significant digits. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Event dummies first stage (2005–2013)

	2005	2006	2007	2008	2009	2010	2011	2012	2013
Expirations X 2005	+***	−***	−***	−***	−***	−***	−***	−***	−***
Expirations X 2006	−**	+***	−***	−***	−***	−***	−***	−***	−***
Expirations X 2007	−**	−***	+***	−***	−***	−***	−***	−***	−***
Expirations X 2008	−**	−***	−***	+***	−***	−***	−***	−***	−***
Expirations X 2009	−*	−***	−***	−***	+***	−***	−***	−***	−***
Expirations X 2010	−**	−***	−***	−***	−**	+***	−***	−***	−***
Expirations X 2011	−**	−***	−***	−***	−**	−***	+***	−***	−***
Expirations X 2012	−**	−***	−***	−***	−*	−***	−***	+***	−***
Expirations X 2013	−**	−***	−***	−**	−*	−**	−***	−***	+***
Expirations X 2014	−**	−***	−***	−**	−*	−**	−***	−***	−***
Expirations X 2015	−**	−***	−***	−**	−*	−***	−***	−***	−***
Expirations X 2016	−*	−***	−***	−**	−*	−***	−***	−***	−***
Expirations X 2017	−	−***	−***	−**	−	−***	−***	−***	−***
Expirations X 2018	−	−***	−***	−**	−	−***	−***	−***	−***
Expirations X 2019	−	−***	−***	−**	−	−***	−***	−***	−***
Expirations X 2020	−	−***	−***	−**	−	−***	−***	−***	−***
Expirations X 2021	−	−***	−***	−**	−	−***	−***	−***	−***
Expirations X 2022	−	−***	−***	−**	−*	−***	−***	−***	−***
Expirations X 2023	−	−***	−***	−***	−*	−***	−***	−***	−***
N	1,134,000	1,134,000	1,134,000	1,134,000	1,134,000	1,134,000	1,134,000	1,134,000	1,134,000
R^2	0.457	0.453	0.454	0.454	0.455	0.458	0.459	0.465	0.467
Outcome mean	0.015	0.015	0.015	0.015	0.015	0.021	0.022	0.022	0.022
F	24.68	23.11	25.08	24.33	26.82	31.73	36.03	42.45	37.53

Notes: To simplify administrative review of results, and reduce the number of values requiring approval, this table presents only the sign (+ or −) and statistical significance of point estimates rather than their exact magnitudes. Statistics at the bottom correspond to observation count, adjusted R^2, mean of the event dummy $\mathbb{1}\{remote_k > \overline{remote_k}\} \times \mathbf{1}\{t\}$ being instrumented, and F-statistic for instrument strength, respectively. All statistics are rounded to 4 significant digits. For space concerns, this table presents the 9 first stages corresponding to event dummies for years 2005 through 2013, with the remaining years shown in Table A8. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Event dummies first stage (2014–2023, excluding 2019)

	2014	2015	2016	2017	2018	2020	2021	2022	2023
Expirations X 2005	−***	−	−	−***	−***	−***	−***	−***	−*
Expirations X 2006	−***	−	−	−**	−***	−***	−***	−***	−
Expirations X 2007	−**	−	−	−**	−***	−***	−***	−**	−
Expirations X 2008	−**	−	−	−*	−***	−***	−***	−**	−
Expirations X 2009	−**	+	−	−*	−***	−***	−***	−***	−
Expirations X 2010	−***	−	+	−*	−***	−***	−***	−***	−
Expirations X 2011	−***	−	+	−**	−***	−***	−***	−***	−
Expirations X 2012	−***	−	+	−**	−***	−***	−***	−***	−
Expirations X 2013	−***	+	+	−**	−***	−***	−***	−***	−
Expirations X 2014	−***	−	−	−**	−***	−***	−***	−***	−
Expirations X 2015	−**	−	−	−**	−***	−***	−***	−**	−
Expirations X 2016	−**	−	−	−*	−***	−***	−***	−**	−
Expirations X 2017	−***	−	−	−**	−***	−***	−***	−***	−
Expirations X 2018	−***	−	−	−*	−***	−***	−***	−***	−
Expirations X 2019	−***	−	−	−*	−***	−***	−***	−***	−
Expirations X 2020	−***	−	−	+***	−***	−***	−***	−***	−
Expirations X 2021	−***	+	−	+	−***	−***	−***	−***	−
Expirations X 2022	−***	+***	−	+	−***	−***	−***	+***	−
Expirations X 2023	−***	−	−	+***	−***	−***	−***	−***	+***
N	1,134,000	1,134,000	1,134,000	1,134,000	1,134,000	1,134,000	1,134,000	1,134,000	1,134,000
R^2	0.468	0.470	0.470	0.468	0.468	0.496	0.496	0.497	0.502
Outcome mean	0.022	0.022	0.022	0.022	0.022	0.030	0.030	0.030	0.029
F	41.95	28.49	30.09	27.66	38.63	34.67	39.43	28.92	24.10

Notes: This table presents the 9 first stages corresponding to event dummies for years 2014 through 2023, excluding 2019, with the remainder shown in Table A7. For other notes, see Table A7. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.